

IDENTIFICATION

PRODUCT CODE: MAINDEC-08-DJKKA-B-D

PRODUCT NAME: PDP-8/A CPU TEST

DATE CREATED: FEBRUARY 15, 1975

MAINTAINER: DIAGNOSTIC GROUP

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LAZI DEP 4000
LA200 SR= 4000 INIT RUN
HALTS A 0222
SET MD → CHECK AC=7777 & LINK IS SET
DISPLAY → PRESS RUN "DO NOT PRESS INIT"
HALT AT 1763 = GOOD PASS.

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HOW TO USE THIS DOCUMENT FOR:

LOADING AND RUNNING THE DIAGNOSTIC

READ SECTION 3, THEN SEE SECTIONS 4.1 AND 4.2 FOR LOADING AND
STARTING INFORMATION. THERE ARE SEVEN (7) PAPER TAPES
ASSOCIATED WITH THIS PROGRAM:

MAINDEC-08-DJKKA-B-PB1	FIELD SERVICE/XOR 4K VERSION
MAINDEC-08-DJKKA-B-PM1	FIELD SERVICE 1K SEGMENT PART 1
MAINDEC-08-DJKKA-B-PM2	FIELD SERVICE 1K SEGMENT PART 2(I/O SIMULATOR REQUIRED)
MAINDEC-08-DJKKA-B-PB2	ACT-8/A 4K VERSION *
MAINDEC-08-DJKKA-B-PB3	ACT-8/A 1K SEGMENT PART 1 *
MAINDEC-08-DJKKA-B-PB4	ACT-8/A 1K SEGMENT PART 2 *
MAINDEC-08-DJKKA-B-PB5	ACT-8/E VERSION **

- * WILL RUN ONLY IN CONJUNCTION WITH THE ACT-8/A INTERFACING
PROGRAM AND THE ACT-8/A OPERATING SYSTEM.
- ** WILL RUN ONLY UNDER CONTROL OF THE ACT-8/E SYSTEM.

FINDING ERROR INFORMATION

SEE SECTION 5 FOR ERROR INFORMATION. CHECK SECTION 3 FOR
POSSIBLE VIOLATION OF PROGRAM RESTRICTIONS.

XOR TROUBLESHOOTING

SEE SECTION 4.2.3 FOR XOR INITIALIZATION INSTRUCTIONS.

1. ABSTRACT

THE PDP-8/A INSTRUCTION TEST IS DESIGNED TO TEST ALL LOGIC ON THE 8/A CPU BOARD THAT IS TESTABLE BY THE USE OF PROGRAMMED INSTRUCTIONS. IN ORDER TO TEST LOGIC ON THE CPU BOARD THAT IS DIRECTLY CONCERNED WITH DATA BREAKS AND/OR INTERRUPTS, A SPECIAL SIMULATOR IS REQUIRED, THIS TEST WILL RUN IN ANY FIELD (SEE 2.2 BELOW), AND WILL POWER FAIL IF A POWER FAIL OPTION IS INSTALLED IN THE CPU.

2. REQUIREMENTS

2.1 HARDWARE

THE FOLLOWING HARDWARE IS REQUIRED FOR THE EXECUTION OF THE 8/A INSTRUCTION TEST:

PROCESSOR-PDP-8/A HEX CPU BOARD (SEE SECTION 3-B)

MEMORY-MINIMUM 1K REQUIRED

OPTIONS- NONE REQUIRED

SPECIAL- PDP-8/A I/O SIMULATOR TO TEST I/O FUNCTIONS.

2.2 STORAGE

THE 8/A INSTRUCTION TEST USES LOCATIONS 0000 THRU 3777. SPECIAL RIM FORMAT BINARY TAPES ARE AVAILABLE TO ALLOW RUNNING THE TEST IN 1K OF MEMORY, BY EXECUTING THE TEST IN TWO CONSECUTIVE 1K SEGMENTS. IN THIS CASE ADDRESSES 0000 THRU 1777 ARE USED, WHEN RUNNING THE XOR VERSION OF THE TEST, LOCATIONS 0000 THRU 4177 ARE USED. THE 4K VERSION OF THIS TEST WILL RUN IN ANY FIELD, SO LONG AS THE LOCATIONS 00000 THRU 00177, AND LOCATION 01777 EXIST AND ARE R/W MEMORY.

2.3 PREREQUISITE SOFTWARE

NO OTHER SOFTWARE IS NECESSARY TO TEST THE 8/A CPU BOARD.

3. RESTRICTIONS

THE FOLLOWING RESTRICTIONS APPLY TO THE 8/A INSTRUCTION TEST:

- A. 1K TO 3K MACHINES-SPECIAL RIM FORMAT PAPER TAPES ARE REQUIRED TO RUN THIS TEST IN MACHINES WITH LESS THAN 4K OF MEMORY. SEE SECTION 4.1.2.
- B. TRADITIONAL PDP-8 COMPUTERS - THIS TEST IS NON-OPERATIONAL ON PDP-8, PDP-8/I, PDP-8/L, PDP-12, AND PDP-8/S COMPUTERS. THIS TEST IS OPERATIONAL ON PDP-8/E, PDP-8/F, AND PDP-8/M COMPUTERS. (SEE SECTION 4.2.1 FOR INITIALIZATION)
- C. INTERRUPTS - NO INTERRUPTS EXCEPT POWER FAIL AND THOSE FROM THE DATA BREAK/INTERRUPT SIMULATOR ARE PERMITTED.

- D. MACHINES WITH PROGRAMMER'S CONSOLE - TO RUN PART 2 OF THIS TEST (USING THE I/O SIMULATOR), IN A MACHINE WITH A PROGRAMMER'S CONSOLE INSTALLED, THE DISPLAY MUST BE SET FOR THE "MD", "STATE", OR "STATUS" OR AN ERROR HALT WILL OCCUR.
- E. DEVICE CODE "77" - IN ORDER TO COMPLETELY TEST ROM H, IT IS NECESSARY TO EXECUTE AN IOT INSTRUCTION WITH BIT 3 OF THE IOT A "1". E.G. 64XX, 65XX, 66XX, 67XX, THE IOT INSTRUCTION USED BY THE TEST IS 6770. IF THIS IOT CONFLICTS WITH A DEVICE ON THE SYSTEM UNDER TEST, DISCONNECT THE DEVICE FROM THE MACHINE WHILE RUNNING PART 2 OF THIS TEST.
- F. IT IS RECOMMENDED THAT THE PROGRAM BE RELOADED AFTER AN ERROR HAS BEEN DETECTED.
- G. NO OTHER DATA BREAK OR INTERRUPT DEVICE MAY BE OPERATING ON THE CPU WHILE RUNNING PART TWO OF THIS TEST WITH THE PDP-8/A I/O SIMULATOR.
- H. IN ORDER TO RUN PART 2 OF THIS TEST IN A CPU THAT CONTAINS A TIME SHARE OPTION(M837 ONLY), THE TIME SHARE OPTION MUST BE DISABLED AS IT MAY CAUSE UNEXPECTED INTERRUPTS.

4. STANDARD TEST PROCEDURE

4.1 PROGRAM LOADING PROCEDURE

4.1.1 4K TO 32K MACHINES

FOR MACHINES WITH 4K TO 32K OF MEMORY THE STANDARD BINARY LOADER TECHNIQUE IS USED. USE PAPER TAPE 08-DJKKA-PB1.

4.1.2 1K TO 3K MACHINES

THE INSTRUCTION TEST IS SEGMENTED INTO TWO (2) 1K SEGMENTS THAT MAY BE RUN CONSECUTIVELY IN A 1K MACHINE. THE 1K SEGMENTS ARE PUNCHED ON TWO RIM FORMAT PAPER TAPES LABELED 08-DJKKA-PM1 AND PM2. SEGMENT 1 SHOULD ALWAYS BE RUN PRIOR TO SEGMENT 2. TO LOAD EITHER SEGMENT PERFORM THE FOLLOWING STEPS:

NOTE: THE SECOND 1K SEGMENT IS ONLY FOR USE WITH AN 8/A I/O SIMULATOR. IF NO SIMULATOR IS AVAILABLE, THE SECOND 1K SEGMENT SHOULD NOT BE RUN.

A. DEPOSIT THE FOLLOWING INSTRUCTIONS INTO PAGE ZERO:

ADDRESS	CONTENTS	
	(HIGH SPEED)	(LOW SPEED)
0156	6014	6032
0157	6011	6031
0160	5357	5357
0161	6016	6036
0162	7106	7106
0163	7006	7006
0164	7510	7510
0165	5374	5357

0166	7006	7006
0167	6011	6031
0170	5367	5367
0171	6016	6034
0172	7420	7420
0173	3776	3776
0174	3376	3376
0175	5357	5356

B. PLACE RIM TAPE IN READER ON LEADER PORTION OF TAPE.

C. LOAD ADDRESS 0156, INIT AND CONTINUE.

D. HALT COMPUTER WHEN ENTIRE TAPE HAS BEEN READ IN.

NOTE: WHEN RUNNING THE SECOND 1K SEGMENT, ADD 2000 TO THE ADDRESS OF ANY ERROR HALTS BEFORE CONSULTING THE LISTING.

4.2 PROGRAM RUN PROCEDURE

4.2.1 INITIALIZATION

IF NO INITIALIZATION IS PERFORMED, THE PROGRAM WILL ASSUME THAT NO PROGRAMMER'S CONSOLE OR I/O SIMULATOR IS AVAILABLE.

PERFORM THE FOLLOWING INITIALIZATION FOR OTHER CONFIGURATIONS:

A. MACHINES WITH PROGRAMMER'S CONSOLE

- 1) LOAD ADDRESS 0021.
- 2) SET SR0=1 TO INDICATE A PROGRAMMER'S CONSOLE IS AVAILABLE.
- 3) SET SR3=1 IF A CPU I/O SIMULATOR IS AVAILABLE.
- 4) SET SR5=1 IF RUNNING THE XOR VERSION.
- 5) SET SR6=1 IF THE PROCESSOR IS A PDP-8/E FAMILY COMPUTER.
(PDP8=E,F, OR M)
- 6) DEPOSIT
- 7) PROGRAM IS NOW INITIALIZED

B. MACHINES WITHOUT PROGRAMMER'S CONSOLE

NOTE: ADDRESS 0021 IS ALREADY INITIALIZED TO INDICATE NO PROGRAMMER'S CONSOLE.

- 1) IF NO PROGRAMMER'S CONSOLE IS AVAILABLE, USE ANY MEANS AVAILABLE TO INITIALIZE LOCATION 21 AS DESCRIBED IN (A.) ABOVE.
- 2) IF RUNNING THE 2K VERSION, OR THE FIRST 1K SEGMENT, INITIALIZE LOCATION 0221 TO A 7000, TO PREVENT THE INITIAL HALT.
- 3) START PROGRAM AT LOCATION 0200.

4.2.2 PROGRAM START

TO START THE TEST, DO THE FOLLOWING:

- A. PERFORM INITIALIZATION IF NECESSARY. SEE SECTION 4.2.1.
- B. LOAD ADDRESS 0200
- C. MAKE SR SETTINGS IF DESIRED. SEE SECTION 6.
- D. DEPRESS INIT, THEN CONTINUE
- E. COMPUTER SHOULD HALT WITH MA=0222, AC=7777, AND LINK=1.

- F. VERIFY THAT THE PC, AC, AND LINK ARE CORRECT. IF THEY ARE CORRECT, PROCEED TO STEP G. IF THEY ARE INCORRECT, REFER TO "ERRORS" SECTION 5.1.
- G. SET FRONT PANEL INDICATE SWITCH TO "MD", "STATE", OR "STATUS" POSITION.
- H. DEPRESS CONTINUE. !!!! DO NOT DEPRESS INIT. !!!!
- I. THE COMPUTER SHOULD RUN CONTINUOUSLY, UNLESS AN ERROR IS DETECTED, OR UNLESS SR3=1, (SEE SECTION 6).

4.2.3 XOR VERSION

THE 2K (STANDARD) VERSION OF THIS TEST CONTAINS ALL NECESSARY CODE TO ALLOW THIS PROGRAM TO RUN ON THE PDP-8/A XOR TESTER. TO RUN THE XOR VERSION, INITIALIZE LOCATION 0021 AS DESCRIBED IN SECTION 4.2.1, THEN START AT LOCATION 0200. AFTER THE FIRST PASS THRU THE DIAGNOSTIC, THE PROGRAM WILL BE MODIFIED FOR XOR TESTING. THE DEVICE CODE SWITCHES ON THE XOR SHOULD BE SET TO 00 WHEN RUNNING THIS TEST.

NOTE: 1K SEGMENTS OF THIS PROGRAM DO NOT CONTAIN XOR CODE.

4.2.4 ACT-8/A VERSION

THE ACT-8/A VERSION OF THIS PROGRAM IS A SPECIALLY MODIFIED VERSION OF THE BASIC PROGRAM. THE DIFFERENCES BETWEEN THE STANDARD VERSION AND THE ACT-8/A VERSION ARE AS FOLLOWS:

- A. ALL ERROR HALTS ARE REPLACED WITH A "JMS SUBROUTINE" IN THE ACT-8/A VERSION. THIS SUBROUTINE GETS THE PC OF THE ERROR AND DOES A JMP TO LOCATION 76520 TO NOTIFY THE ACT-8/A MONITOR OF THE PROGRAM DETECTED ERROR.
- B. AT THE END OF EACH SUCCESSFUL PASS OF THE TEST, THE PROGRAM DOES A JMS TO LOCATION 76500 TO NOTIFY THE ACT-8/A MONITOR OF THE SUCCESSFUL PASS. RETURN IS MADE TO THE PROGRAM FROM THE MONITOR AND TESTING RESUMES. TESTING WILL CONTINUE WITH THE MONITOR BEING NOTIFIED OF EACH SUCCESSFUL PASS UNTIL EITHER AN ERROR OCCURS OR THE MONITOR LOADS ANOTHER PROGRAM.
- C. XOR CODE IS NOT IMPLEMENTED IN THE ACT-8/A VERSION.
- D. THE ACT-8/A VERSION HAS THREE BINARY TAPES:

MD-08-DJKKA-PB2 4K VERSION
MD-08-DJKKA-PB3 1K SEGMENT PART 1
MD-08-DJKKA-PB4 1K SEGMENT PART 2

THESE THREE PROGRAMS ARE THE EQUIVALENT OF THE THREE FIELD SERVICE VERSIONS, WITH THE EXCEPTION OF THE DIFFERENCES NOTED ABOVE.

4.2.5 ACT-8/E VERSION

THE ACT-8/E VERSION OF THIS TEST IS A SPECIALLY MODIFIED VERSION OF THE BASIC 2K PROGRAM. USE PAPER TAPE DJKKA-PB5. THE DIFFERENCES BETWEEN THE ACT-8/E VERSION AND THE BASIC 2K VERSION ARE AS FOLLOWS:

- A. ALL ERROR HALTS ARE REPLACED WITH A "JMS SUBROUTINE" IN THE ACT-8/E VERSION. THE SUBROUTINE NOTIFIES THE ACT-8/E MONITOR

OF THE ERROR.

- B. IF BIT2 OF LOCATION 0022 = 1, THEN AFTER ONE PASS THROUGH THE TEST THE ACT-8/E MONITOR IS NOTIFIED OF THE GOOD PASS AND PROGRAM EXECUTION CEASES.
- C. IF BIT2 OF LOCATION 0022 = 0, THE PROGRAM RUNS FOR 10 MINUTES (AS TIMED IN A CORE MEMORY MACHINE), AT THE END OF TEN MINUTES THE ACT-8/E MONITOR IS NOTIFIED OF A GOOD PASS.
- D. XOR CODE IS NOT IMPLEMENTED IN THE ACT-8/E VERSION.
- E. POWER FAIL IS NOT IMPLEMENTED IN THE ACT-8/E VERSION.
- F. THE ACT-8/E VERSION RUNS ONLY IN FIELD 0.

5. ERRORS -----

5.1 ERROR HALTS -----

ALL PROGRAM ERRORS ARE INDICATED BY MEANS OF ERROR HALTS. THE PROGRAM LISTING CONTAINS A BRIEF EXPLANATION OF THE ERROR TO THE RIGHT OF EACH HALT IN THE LISTING. USE THE PC CONTENTS AFTER THE ERROR HALT TO FIND THE ERROR INFORMATION IN THE LISTING.

NOTE: IF RUNNING THE 2ND 1K SEGMENT IN A 1K CPU, ADD 2000 TO THE ADDRESS OF ANY ERROR HALT BEFORE CONSULTING THE LISTING.

THE FOLLOWING ERRORS HAVE BEEN INCLUDED HERE, BECAUSE THEY REQUIRE FURTHER EXPLANATION:

PC=0036 THIS INDICATES THAT AN UNEXPECTED INTERRUPT WAS RECEIVED BY THE CPU, LOCATION 0000 WILL CONTAIN THE ADDRESS+1 OF WHERE THE PROGRAM WAS INTERRUPTED. E.G. IF LOCATION 0000 CONTAINS 2635, THEN THE COMPUTER WAS INTERRUPTED AFTER THE INSTRUCTION AT LOCATION 2634.

PC=0221 THIS IS NOT AN ERROR HALT UNLESS THE AC IS NOT EQUAL TO 7777 OR THE LINK IS NOT EQUAL TO 1. IF THE AC OR THE LINK IS INCORRECT, LOAD ADDRESS 0200, DEPRESS THE HALT KEY, AND DEPRESS CLEAR MOMENTARILY. THE PROGRAM MAY NOW BE EXECUTED ONE INSTRUCTION AT A TIME BY DEPRESSING THE CONTINUE KEY. THE OPERATOR SHOULD CHECK THE CONTENTS OF THE AC AND LINK AGAINST THE EXPECTED CONTENTS GIVEN IN THE PROGRAM LISTING AFTER EXECUTING EACH INSTRUCTION. WHEN AN INSTRUCTION IS EXECUTED AND THE COMPUTER REGISTERS NO LONGER AGREE, THE FAILING INSTRUCTION HAS BEEN FOUND.

PC=1631 A SKIP ERROR OCCURED DURING THE TEST OF ROMS "D" AND "F". THE INSTRUCTION IN THE AC WAS EXECUTED, AND RESULTED IN A SKIP WHEN NONE WAS EXPECTED, OR DID NOT SKIP WHEN IT WAS EXPECTED TO SKIP. MAKE A NOTE OF THE INSTRUCTION, THEN DEPRESS CONTINUE TO GET THE CONTENTS OF THE AC, MQ AND LINK AT THE TIME OF THE FAILURE. TO EXECUTE THE FAILING INSTRUCTION AGAIN, DEPRESS CONTINUE. TO EXECUTE THE NEXT INSTRUCTION (OR NEXT DATA PATTERN WITH SAME INSTRUCTION), MAKE A NOTE OF THE AC, LINK, AND MQ CONTENTS FOR REFERENCE, THEN LOAD ADDRESS 1673, DEPRESS

CLEAR, THEN CONTINUE. IF FURTHER ERROR HALTS OCCUR, THE ERROR INFORMATION SHOULD BE RECORDED FOR USE IN DETERMINING A PATTERN FOR THE FAILURE, (E.G. CLL CML COMBINATION SKIPS, SPA SNA WON'T SKIP IF LINK IS SET, ETC.)

PC=1650 A DATA ERROR OCCURRED DURING THE TEST OF ROMS "D" AND "F". THE INSTRUCTION IN THE AC WAS EXECUTED, AND RESULTED IN INCORRECT CONTENTS OF THE AC, MQ, OR LINK. MAKE A NOTE OF THE INSTRUCTION, THEN DEPRESS CONTINUE TO GET THE EXPECTED CONTENTS OF THE AC, LINK, AND MQ. MAKE A NOTE OF THE EXPECTED CONTENTS, THEN DEPRESS CONTINUE TO GET THE CONTENTS OF THE AC, MQ, AND LINK AS THEY WERE FOUND AFTER THE INSTRUCTION WAS EXECUTED. TO EXECUTE SAME INSTRUCTION AGAIN DEPRESS CONTINUE. TO EXECUTE NEXT INSTRUCTION, OR NEXT DATA PATTERN FOR SAME INSTRUCTION, LOAD ADDRESS 1673, CLEAR AND CONTINUE. IF FURTHER ERRORS OCCUR, THE ERROR INFORMATION SHOULD BE RECORDED FOR USE IN FINDING A POSSIBLE PATTERN IN THE ERROR. E.G. CLL CMA CML COMBINATION DOES NOT WORK CORRECTLY.

PC=ALL OTHER HALTS REFER TO PROGRAM LISTING UNDER PROPER PC.

5.1.2 LOOPING ON ERROR

NOTE: DISREGARD THIS INFORMATION WHEN RUNNING THE XOR VERSION.
XOR LOOPING IS AUTOMATIC.

TO LOOP ON A FAILING INSTRUCTION, (OTHER THAN ROM "D" & "F" TEST), IT IS NECESSARY TO DEPOSIT A JUMP INSTRUCTION IN PLACE OF THE ERROR HALT THAT IS OCCURRING. THE JUMP INSTRUCTION SHOULD CAUSE THE PROGRAM TO JUMP BACK TO THE POINT WHERE THE FAILING INSTRUCTION IS EXECUTED. NOTE: IF SPECIAL CONDITIONS ARE REQUIRED (E.G. AC & LINK MUST BE CLEAR, AC MUST BE EQUAL TO 7777, ETC) THE OPERATOR WILL HAVE TO DEPOSIT THE PROPER INSTRUCTIONS TO CAUSE THESE CONDITIONS PREVIOUS TO THE FAILING INSTRUCTION, AND MAKE THE JMP AFTER THE FAILING INSTRUCTION JUMP ACCORDINGLY.

EXAMPLE:

ADDRESS	CONTENTS	MNEMONIC	
0361	1024	TAD K1	/AC TO 0001 LINK=1
0362	1054	TAD K7777	/AC TO 0000 LINK TO 0
0363	7450	SNA	/SHOULD NOT SKIP IF AC=0000
0364	7430	SZL	/SHOULD SKIP IF LINK=0
0365	7402	HLT	/CARRY FAILED TO PROPAGATE THRU ADDER

IF THE PROGRAM IS HALTING AT ADDRESS 0365 WITH THE AC NON-ZERO, ONE OR BOTH OF THE TAD INSTRUCTIONS ARE FAILING. IN ORDER TO LOOP ON THE FAILING INSTRUCTIONS, A JUMP TO THE FIRST TAD INSTRUCTION (LOCATION 0361) COULD BE PLACED AT LOCATION 0365, BUT THE AC WOULD NOT BE CLEAR, AND THE LINK WOULD NOT BE SET WHEN THE FIRST TAD IS EXECUTED. IN ORDER TO LOOP CORRECTLY, THE FOLLOWING PATCH IS REQUIRED TO SET THE AC AND LINK TO THE PROPER VALUES. "*" INDICATES INSTRUCTIONS THAT WERE DEPOSITED FOR LOOPING PURPOSES.

ADDRESS	CONTENTS	MNEMONIC	
0360*	7320	CLA CLL CML	(CLEAR AC, SET LINK)
0361	1024	TAD K1	
0362	1054	TAD K7777	
0363	7450	SNA	
0364	7430	SZL	
0365*	5360	JMP ,-5	(FAILED, DO AGAIN)
0366*	7402	HLT	(DID NOT FAIL)

THE LOOP SHOULD BE EXECUTED THE FIRST TIME DOING ONE INSTRUCTION AT A TIME WITH HALT/SS SELECTED, TO INSURE THAT THE INSTRUCTION IS STILL FAILING, AND THAT ANY INSTRUCTIONS INSERTED (IN THE ABOVE EXAMPLE THE JMP ,-5, AND THE CLA CLL CML) ARE NOT ALSO FAILING.

IN ORDER TO RUN THE PROGRAM AGAIN AFTER REPAIRS HAVE BEEN MADE, A RELOAD OF THE PROGRAM IS REQUIRED.

5.2 ERROR PRINTOUTS

NO ERROR PRINTOUTS ARE USED BY THIS TEST, ALL ERRORS ARE INDICATED BY PROGRAM HALTS.

6. SWITCH REGISTER SETTINGS

6.1 NORMAL OPERATING SWITCHES

MACHINES WITH PROGRAMMER'S CONSOLE SHOULD INITIALIZE THE PROGRAM TO USE THE PROGRAMMER'S CONSOLE SWITCH REGISTER, (SEE SECTION 4.2.1). MACHINES WITHOUT PROGRAMMER'S CONSOLE USE LOCATION 0020 IN LIEU OF A SWITCH REGISTER. REGARDLESS OF WHETHER THE PROGRAMMER'S CONSOLE SWITCH REGISTER OR ADDRESS 0020 IS USED, THE BITS HAVE THE FOLLOWING PURPOSES:

BIT	FUNCTION WHEN 0	FUNCTION WHEN 1
0-2	NOT USED	NOT USED
3	LOOP ON COMPLETE TEST	HALT AT END OF TEST
4-11	NOT USED	NOT USED

6.2 ERROR SWITCHES

NO ERROR SWITCHES ARE PROVIDED, SEE SECTION 5.1.2 FOR ERROR LOOPING PROCEDURE

7. REVISIONS

THE "B" REVISION WAS RELEASED TO DELETE THE "LOAD ADDRESS" TEST WHICH COULD CAUSE MARGINAL (BUT LEGAL) FAILURES, CERTAIN TESTS WERE ALSO ADDED TO CATCH FAILURES THAT WERE NOT CAUGHT IN THE FAULT INSERTION OF THE "A" REVISION. THE "B" VERSION OF THE SOURCE ALSO SUPPORTS AN ACT-8/E VERSION.

8. PROGRAM DESCRIPTION

8.1 BASIC INSTRUCTION TESTS

DURING THE FIRST PORTION OF THE TEST, ALL BASIC GROUP 1 AND GROUP 2 OPERATE INSTRUCTIONS ARE TESTED, NO COMBINED OPERATES ARE TESTED EXCEPT CLA CLL, THEN ALL MRI INSTRUCTIONS ARE TESTED, USING BOTH DIRECT AND INDIRECT ADDRESSING, DURING THIS SECTION THE ADDER IS TESTED BY THE USE OF TAD INSTRUCTIONS AND IAC, FINALLY BASIC GROUP 3 OPERATE INSTRUCTIONS ARE TESTED.

8.2 TEST OF ROMS "D" AND "F"

ROMS "D" AND "F" ON THE CPU MODULE ARE TESTED BY EXECUTING ALL GROUP 1,2, AND 3 OPERATE INSTRUCTIONS OF THE FORM: 7XX0, 7XX1. FOR EACH INSTRUCTION THUS TESTED, 8 DIFFERENT DATA COMBINATIONS OF AC,MQ, AND LINK ARE USED, THE SEQUENCE OF ROM TESTING IS AS FOLLOWS:

- A. THE AC, LINK, AND MQ ARE SET TO SPECIFIED VALUES.
- B. THE INSTRUCTION TO BE TESTED IS EXECUTED.
- C. THE AC, LINK, AND MQ ARE SAVED, AND WHETHER THE INSTRUCTION SKIPPED IS NOTED.
- D. THE AC, LINK, AND MQ ARE SET TO THE SAME VALUES AS IN STEP A.
- E. THE INSTRUCTION IS SIMULATED USING ONLY THOSE INSTRUCTIONS THAT WERE TESTED DURING THE FIRST PART OF THE TEST, (BASIC OPERATE AND MRI INSTRUCTIONS)
- F. THE RESULTS OF THE SIMULATION ARE COMPARED TO THE RESULTS OF THE ACTUAL INSTRUCTION, ANY DIFFERENCES RESULT IN AN ERROR HALT.

8.3 DATA BREAK/INTERRUPT TESTING

THE LOGIC ON THE CPU BOARD CONCERNED WITH DATA BREAKS AND INTERRUPTS IS TESTED BY THE USE OF A SPECIAL SIMULATOR, I/O SKIPS, AC TRANSFERS, INTERRUPTS, INDICATE LOGIC, AND DATA BREAKS ARE TESTED IF THE SIMULATOR IS AVAILABLE, ALL OMNIBUS LINES ARE TESTED EITHER DIRECTLY OR INDIRECTLY WITH THE EXCEPTION OF "NEXT TIME STATE STALL" (BR2) AND THE UNUSED OMNIBUS LINE(BS2),

8.4 UNTESTED LOGIC

DUE TO CERTAIN HARDWARE RESTRICTIONS, SOME LOGIC ON THE POP-8/A CPU BOARD IS NOT TESTED BY THIS PROGRAM, BELOW IS A LIST OF LOGIC THAT IS KNOWN TO BE UNTESTED,

ROM A - ADDRESSES 00 THRU 03 (EXTENDED LOAD ADDRESS 7)
ADDRESSES 04 THRU 07 (EXTENDED LOAD ADDRESS)
ADDRESSES 14 THRU 17 (LOA

/8A CPU TEST 08-DJKKA=B
 /PROGRAMMER: MIKE WARE
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 /*****

/ASSEMBLY INSTRUCTIONS:

/1. IF ASSEMBLING 2K VERSION, INSERT THE FOLLOWING: TWOK=0
 /2. IF ASSEMBLING THE FIRST 1K SEGMENT, INSERT THE FOLLOWING: ONEKP1=0
 /3. IF ASSEMBLING THE SECOND 1K SEGMENT, INSERT THE FOLLOWING: ONEKP2=0
 / ONE AND ONLY ONE OF THE ABOVE THREE DEFINITIONS MUST BE INSERTED IN
 / THIS SOURCE BEFORE ANY OTHER ASSEMBLY CODE.
 /4. IF ASSEMBLING ANY OF THE ABOVE 3 VERSIONS FOR USE ON THE ACT-8/A LINE,
 / INSERT ONE OF THE ABOVE DEFINITIONS AND THE FOLLOWING: ACT8A=0
 /5. IF ASSEMBLING THE ACT-8/E VERSION, DEFINE ACT8E=0 AND ACT8A=0,
 / THE ACT-8/E VERSION MUST BE A 2K VERSION, 1K VERSIONS ARE NOT
 / SUPPORTED ON THE ACT-8/E LINE.
 /*****

0000 TWOK=0
 /INSTRUCTION EQUALITIES
 7402 HLT=7402 /HALT
 7002 BSW=7002 /BYTE SWAP
 7421 MQL=7421 /AC TO MQ, 0 TO AC
 7501 MQA=7501 /MQ + AC TO AC
 7621 CAM=7621 /CLEAR AC AND MQ
 7521 SWP=7521 /SWAP AC AND MQ
 7701 ACL=7701 /MQ TO AC
 6214 RDF=6214 /READ DATA FIELD
 6224 RIF=6224 /READ INSTRUCTION FIELD
 6000 SKON=6000 /SKIP IF INTERRUPT ON, TURN INTERRUPT OFF
 6001 ION=6001 /TURN INTERRUPT ON
 6002 IOF=6002 /TURN INTERRUPT OFF
 6003 SRQ=6003 /SKIP ON INTERRUPT REQUEST
 6004 GTF=6004 /GET FLAGS
 6005 RTF=6005 /RESTORE FLAGS
 6006 SGT=6006 /SKIP ON "GREATER THAN" FLAG, NOTE: THE "GT" FLAG
 /IS NOT IMPLEMENTED IN THE PDP-8/A
 6007 CAF=6007 /CLEAR ALL FLAGS, AND CLEAR AC AND LINK
 6102 SPL=6102 /SKIP ON POWER LOW
 /*****

/XOR INSTRUCTIONS

6170 XRON=6170 /TURN XOR ON
 6171 SKXR=6171 /SKIP ON XOR ERROR 1
 6172 XRCI=6172 /TURN OFF XOR INTERRUPT
 6173 STIP=6173 /SKIP IF MUT ON AND FIRST XRON DONE
 6174 XRSI=6174 /SET XOR INTERRUPT ENABLE
 6175 SXRC=6175 /SKIP ON XOR ERROR 2
 6176 XRT0=6176 /SET TIME OUT FLOP
 /*****

/8-A I/O SIMULATOR INSTRUCTIONS

6140 CNTENA=6140 /START TP1 COUNTER, ENABLE STATUS READING VIA 6141 IF BIT 11=1
 6141 READA=6141 /READ REGISTER A, OR DATA BREAK DATA, OR STATUS.
 6142 STROB=6142 /GATE C LINES AS DEFINED BY REGISTER B TO CPU
 6143 SETBK=6143 /SET BREAK REQUEST, (DELAY MUST ALSO TIME OUT BEFORE BREAK OCCURS)
 6144 SKPOFV=6144 /SKIP IF OVERFLOW F/F SET IN SIMULATOR, CLEARS SIMULATOR INTERRUPT.
 6145 CLRAL=6145 /CLEAR SIMULATOR LOGIC.
 6146 LOADA=6146 /LOAD REGISTER A WITH AC.
 6147 LOADB=6147 /LOAD REGISTER B WITH AC.
 /*****

0000 *0 HLT/HLT(7402) /A HALT HERE INDICATES A JUMP'S FAILURE TO GATE MD TO PC,
 0001 7402 HLT/RMF(6244) /CHANGE IF AND OF BACK
 0002 7402 HLT/JMP SKPCHN /GET BACK TO CORRECT FIELD
 0003 7777 BIT3, 7777
 0004 0000 BIT4, 0
 0005 0000 BIT5, 0
 0006 0000 BIT6, 0
 0007 0000 0
 0010 0000 0
 0011 0000 0
 0013 *13 0
 0014 0000 0
 0015 0000 0
 0016 0000 0
 0017 0000 0
 /*****

/PDP-8 STANDARDIZED SWITCHES AND HARDWARE DESIGNATOR WORDS

0020 *20 0
 0020 0000 SWITCH, 0 /PSEUDO SWITCH REGISTER
 0021 0000 HWRDES, 0 /BIT3=1 FOR I/O TESTER, BIT5=1 FOR XOR
 /BIT6=1 FOR 8/E TYPE CPU
 0022 0000 0
 /*****
 0023 3074 SKPCHN, DCA SAVAC /SAVE THE AC FOR A MOMENT
 0024 6201 CDF 00 /DATA FIELD TO 0 FOR GETTING INTERRUPT PC
 0025 1440 TAD I K0 /GET CONTENTS OF 00000,
 0026 3000 DCA 0 /PUT INTERRUPT PC IN THIS FIELD
 0027 6102 SPL /POWER FAILURE?
 0030 7410 SKP /NO, CHECK FOR XOR INTERRUPT
 0031 5125 JMP POWRON /YES, PREPARE FOR BLACKOUT
 0032 6244 RMF /RESTORE OF
 0033 6144 SKPOFV /CLEAR SIMULATOR INTERRUPT
 0034 7000 K7000, NOP
 0035 6175 SXRC /XOR ERROR? (TYPE 2)
 0036 7402 CHNCON, HLT /NO, UNEXPECTED INTERRUPT, REPLACED BY A JMP I 7
 /IF SOME OTHER INTERRUPT EXPECTED,
 0037 5166 JMP XORBAK /YES, PROCESS XOR INTERRUPT.
 /*****

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/CONSTANTS
0040 0000 K0, 0000
0041 0001 K1, 0001
0042 0002 K2, 0002
0043 0004 K4, 0004
0044 0010 K10, 0010
0045 0020 K20, 0020
0046 0040 K40, 0040
0047 0077 K77, 0077
0050 0100 K100, 0100
0051 0200 K200, 0200
0052 0400 K400, 0400
0053 1111 K1111, 1111
0054 1777 K1777, 1777
0055 2000 K2000, 2000
0056 2525 K2525, 2525
0057 2552 K2552, 2552
0060 3333 K3333, 3333
0061 4000 K4000, 4000
0062 4444 K4444, 4444
0063 5225 K5225, 5225
0064 5252 K5252, 5252
0065 5253 K5253, 5253
0066 6666 K6666, 6666
0067 7700 K7700, 7700
0070 7721 K7721, 7721
0071 7770 K7770, 7770
0072 7777 K7777, 7777
IFNDEF ACT8E <

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/*****
/SCRATCH LOCATIONS

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0074 0000 SAVAC, 0 /POWERFAIL AC STORAGE
0075 0000 SAVFLG, 0 /POWERFAIL FLAG STORAGE
0076 0000 SAVMQ, 0 /POWERFAIL MQ STORAGE
0077 0000 SAVRET, 0 /POWERFAIL RETURN STORAGE
0100 0000 PARTWO, 0 /CONTAINS 7777 WHEN IN I/O SIM PORTION OF CPU TEST
0101 0000 ACWAS, 0
0102 0000 MQWAS, 0
0103 0000 LKWAS, 0
0104 0000 SKPPED, 0
0105 0000 SOMSKP, 0
0106 0000 SAVFLD, 0 /STORAGE FOR IF AND DF
0117 BIT6=POINTD
0117 BIT7=0007
0112 BIT11=TESLOC
0107 0000 ACDATA, 0
0110 0000 MQDATA, 0
0111 0000 LKDATA, 0
0112 0000 TESLOC, 0
0113 0114 POINTR, +4
0114 2526
0115 0116 POINTB, POINTC

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0116 0000 POINTC, 0
0117 0000 POINTD, 0
0120 1310 POINTE, JMSLOC-1
0121 1324 POINTF, JMSLOC-2
0122 1372 INSTTR, INSTRT
0123 1577 TEST, TESTS
0124 1745 CKSWIT, XOR29> /CONTAINS THE CONSTANT 0200 FOR PART 2 OF THE 1K VERSION.

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/*****
/POWER FAIL ROUTINE
0125 6004 POWRDN, GTF /GET LINK AND FLAGS
0126 3536 DCA I PAVFLG /SAVE FLAGS
0127 7521 SWP: /MQ TO AC
0130 3537 DCA I PSVMQ /SAVE MQ CONTENTS
0131 1000 TAD 0 /GET RETURN ADDRESS FOR POWER-UP
0132 3540 DCA I PAVRET /SAVE RETURN ADDRESS IN FLD 0,
0133 1073 TAD RETINS /INSTRUCTION FOR POWER-UP EXECUTION
0134 3440 DCA I K0 /PUT IT IN ADDRESS 00000
0135 7402 KSTOP, HLT /WAIT OUT POWER FAILURE
0136 0075 PAVFLG, SAVFLG
0137 0076 PSVMQ, SAVMQ
0140 0077 PAVRET, SAVRET
0141 5524 TWOBK, JMP I CKSWIT
0142 PWRADD,
0155 +13

/POWER UP ROUTINE
/PWRUP, TAD SAVMQ
/ MQ: /RESTORE MQ
/ TAD SAVFLG /RESTORE FLAGS
/ RTF:
/ CLA
/ TAD I PPRTWO
/ SEA CLA /IN THE SECOND PART OF THE TEST?
/ JMP: TWOBK /YES, BEGIN SECOND PART OVER,
/ TAD SAVAC /RESTORE AC
/ JMP: I SAVRET /RETURN TO PROGRAM
/PPRTWO, PARTWO

/*****
/XOR CODE
0156 +156
/THIS ROUTINE MUST ALWAYS REMAIN AT LOCATION 0156. IT IS REPLACED
/BY THE RIM LOADER PROGRAMS IN 1K MACHINES, AND BY ACT-8/A CODE ON THE ACT LINE.
/ROUTINE FOR SAVING XOR RETURN FOR ERROR LOOPING
0156 0000 POINT, 0
0157 7300 CLA CLL
0160 6171 SKXR /TYPE 1 ERROR EXISTING?
0161 7410 SKP: /NO, SWAP POINTER AND GO NEXT TEST
0162 5166 JMP: XORBK /ERROR EXISTING, LOOP ON LAST TEST,
0163 1156 TAD POINT /SWAP POINTER
0164 3173 DCA POINTX
0165 5556 JMP: I POINT

/XOR RETURN AND INITIALIZE
0166 6007 XORBK, CAF /CLEAR AC, LINK, AND WORLD

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0167 7621 CAM /CLEAR MQ
0170 6145 CLRAL /CLEAR I/O SIMULATOR
0171 6001 ION
0172 5573 JMP I POINTX /RETURN TO TEST THAT ERRORED.
0173 0174 POINTX, POINTX
0174 7402 POINTX, HLT /XOR INTERRUPT WHEN NOT RUNNING XOR VERSION
0177 0177 *177
0177 7770 DATPAT, 7770

0200 *200
/*****
/INITIAL CONDITIONS: AC AND LINK CLEAR FROM INITIALIZE
/*****
/TEST ALL BASIC SKIPS TO EITHER SKIP OR NOT SKIP WHEN AC=7777.
START, CLA/JMS POINT FOR XOR

0200 7200 SNA
0201 7450 SEL
0202 7430 HLT /SNA SKIPS WHEN AC CLEAR, OR SEL DOES NOT SKIP WHEN LINK = 0
0203 7402 CMA /AC TO 7777
0204 7040 ION /ENABLE INTERRUPTS FOR POSSIBLE POWER FAIL
0205 6001 SNA
0206 7450 HLT /CMA SKIPPED OR DID NOT COMPLEMENT, OR SNA DID NOT SKIP, OR ION SKIPPED
0207 7402 SEA
0210 7440 SKP
0211 7410 HLT /SNA CLEARED AC, OR SEA SKIPPED, OR SKP FAILED
0212 7402 SMA
0213 7500 HLT /SEA OR SKP CLEARED AC0, OR SMA FAILED TO SKIP
0214 7402 SPA
0215 7510 SKP
0216 7410 HLT /SMA CLEARED AC0, OR SPA SKIPPED WHEN AC0=1
0217 7402 CML /SET LINK TO 1
0220 7020 EXHLT1, HLT/JMS POINT FOR XOR /EXPECTED HALT #1. AC SHOULD EQUAL 7777, LINK=1
0221 7402 /VERIFY THAT AC=7777 AND LINK=1, THEN CONTINUE
/*****
/TEST CLA CLL TO CLEAR AC AND LINK
/*****
0222 7300 TSCACL, CLA CLL
0223 7450 SNA
0224 7430 SEL
0225 7402 HLT /CLA CLL DID NOT CLEAR AC OR LINK, OR SNA SKIPPED
/WHEN AC=0000, OR SEL DID NOT SKIP WHEN LINK=0
0226 7040 CMA /AC TO 7777
/*****
/TEST BASIC SKIPS TO SKIP OR NOT SKIP WHEN AC=0000
/*****
0227 7040 TSBSSK, CMA /AC TO 0000
0230 7440 SEA
0231 7402 HLT /CMA DID NOT COMPLEMENT AC OR SEA FAILED TO SKIP, OR CMA SKIPPED
0232 7510 SPA
0233 7402 HLT /SPA FAILED TO SKIP WHEN AC0=0
0234 7500 SMA
0235 7410 SKP
0236 7402 HLT /SMA SKIPPED WHEN AC0=0
/*****
/TEST IAC TO INCREMENT AC TO 0000 AND SET LINK
/*****

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0237 7100 TSTIAC, CLL /THIS INSTRUCTION NOT YET TESTED
0240 7040 CMA /AC TO 7777
0241 7001 IAC /AC TO 0000, LINK TO 1
0242 7440 SEA
0243 7402 HLT /SEA FAILED TO SKIP, OR IAC DID NOT INCREMENT AC TO 0000
/*****
/TEST TO SEE IF LINK COMPLEMENTED TO A 1 ON A CARRY OUT OF ADDER
/*****
0244 7420 TSTLCM, SNL
0245 7402 HLT /LINK DID NOT COMPLEMENT ON CARRY OUT, OR CLL FAILED
/OR SNL FAILED TO SKIP FOR LINK = 1
0246 7430 SEL
0247 7410 SKP
0250 7402 HLT /SEL SKIPPED ON LINK=1, OR SNL CLEARED LINK, OR SKIP FAILS WHEN LINK =1
/*****
/TEST CLL TO CLEAR LINK
/*****
0251 7100 TSTCLL, CLL
0252 7420 SNL
0253 7410 SKP
0254 7402 HLT /CLL FAILED, OR SNL SKIPPED WHEN LINK=0, OR SKP FAILED WHEN LINK=0
0255 7430 SEL
0256 7402 HLT /SEL FAILED TO SKIP, OR SNL OR SKP SET LINK
/*****
/TEST ABILITY OF CML TO SET LINK
/*****
0257 7020 TSTCML, CML /LINK TO 1
0260 7420 SNL
0261 7402 HLT /CML DID NOT SET LINK
/*****
/TEST ABILITY OF LINK TO COMPLEMENT FROM A 1 TO A 0 ON A CARRY OUT
/*****
0262 7040 CMA /AC TO 7777, LINK=1
0263 7001 IAC /AC TO 0000, CARRY TO LINK, LINK TO 0
0264 7430 SEL
0265 7402 HLT /CARRY OUT DID NOT COMPLEMENT LINK TO A 0
/*****
/TEST ABILITY OF CML TO COMPLEMENT LINK FROM A 0 TO A 1 AND BACK TO A 0
/*****
0266 7200 XOR03, CLA/JMS POINT FOR XOR
0267 7020 CML /LINK TO 1
0270 7020 CML /LINK TO 0
0271 7430 SEL
0272 7402 HLT /CML DID NOT COMPLEMENT LINK FROM A 1 TO A 0
0273 7440 SEA
0274 7402 HLT /CML CHANGED AC
/*****
/TEST CLA TO CLEAR AC AND NOT CLEAR LINK
/*****
0275 7020 CML /MAKE LINK A 1
0276 7040 CMA /AC TO 7777
0277 7200 CLA /AC TO 0000
0300 7420 SNL
0301 7402 HLT /CLA CLEARED LINK
0302 7440 SEA
0303 7402 HLT /CLA DID NOT CLEAR AC
/*****
/TEST NOP TO NOT CHANGE AC OR LINK
/*****
0304 7000 NOP /AC=0000 LINK=1
0305 7420 SNL
0306 7402 HLT /NOP SKIPPED OR CLEARED LINK

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0307 7440 SZA
0310 7402 HLT /NOP SET AC BIT
0311 7040 CMA /AC TO 7777
0312 7020 CML /LINK TO 0
0313 7000 NOP /AC=7777, LINK=0
0314 7430 SEL
0315 7402 HLT /NOP SKIPPED OR SET LINK
0316 7450 SNA
0317 7402 HLT /NOP CLEARED AC
/*****
/TEST RAL TO NOT PICK UP BITS BY SHIFTING ZEROES
0320 7000 XOR04, NOP/JMS POINT FOR XOR
0321 7200 CLA /AC TO 0000
0322 7004 RAL
0323 7450 SNA
0324 7430 SEL
0325 7402 HLT /RAL OF ZEROES PICKED UP AC BIT OR LINK BIT
/*****
/TEST RTL TO NOT PICK UP BITS WHEN SHIFTING ALL ZEROES
0326 7006 RTL
0327 7450 SNA
0330 7430 SEL
0331 7402 HLT /RTL PICKED UP BIT IN AC OR LINK WHEN SHIFTING ZEROES
/*****
/TEST RAR TO NOT PICK UP BITS WHEN SHIFTING ALL ZEROES
0332 7010 RAR
0333 7450 SNA
0334 7430 SEL
0335 7402 HLT /RAR PICKED UP BIT IN AC OR LINK WHEN SHIFTING ZEROES
/*****
/TEST RTR TO NOT PICK UP BITS WHEN SHIFTING ALL ZEROES
0336 7012 RTR
0337 7450 SNA
0340 7430 SEL
0341 7402 HLT /RTR PICKED UP BIT IN AC OR LINK WHEN SHIFTING ZEROES
/*****
/TEST BSW TO NOT PICK UP ANY BITS WHEN SWAPPING ZEROES, AND TO NOT AFFECT LINK
0342 7002 BSW
0343 7450 SNA
0344 7430 SEL
0345 7402 HLT /BSW PICKED UP BIT IN AC WHEN SWAPPING ALL ZEROES, OR SET LINK
/*****
/FIRST TEST OF MRI
/*****
/TEST TAD TO ADD 7777 TO A CLEAR AC
0346 7000 XOR05, NOP/JMS POINT FOR XOR
0347 1072 TAD K7777 /AC TO 7777, LINK=0
0350 7450 SNA
0351 7402 HLT /TAD DID NOT LOAD AC, OR TAD SKIPPED,
0352 7430 SEL
0353 7402 HLT /TAD SET LINK WHEN NO CARRY OUT EXPECTED
/*****
/TEST TAD TO ADD 1 TO AC=7777 TO PRODUCE AC=0000 AND LINK=1
0354 1041 TAD K1 /AC TO 0000, LINK TO 1
0355 7440 SZA

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0356 7402 HLT /TAD USED INCORRECT VALUE, OR ADDER CARRY CKT
/FAULTY OR TAD SKIPPED
0357 7420 SNL
0360 7402 HLT /CARRY OUT OF ADDER DID NOT COMPLEMENT LINK
/*****
/TEST ADDER CARRY STRUCTURE TO GENERATE CARRY THROUGH ALL POSITIONS
/UNDER OPPOSITE CONDITIONS FROM PREVIOUS TEST
0361 1041 TAD K1 /AC TO 0001, LINK=1
0362 1072 TAD K7777 /AC TO 0000, LINK TO 0
0363 7450 SNA
0364 7430 SEL
0365 7402 HLT /CARRY FAILED TO PROPAGATE ALL THROUGH ADDER TO LINK
/*****
/TEST ABILITY TO ADD 0000 TO A CLEAR AC TO PRODUCE A CLEAR AC
0366 1040 TAD K0 /AC=0000, LINK=0
0367 7450 SNA
0370 7430 SEL
0371 7402 HLT /ADDING 0000 TO 0000 PRODUCED NON-ZERO RESULT
/OR COMPLEMENTED LINK
/*****
/TEST ADDER'S ABILITY TO PROPAGATE CARRY WHEN ACN=1, MDN=1, AND CARRY IN N=1
0372 1072 TAD K7777 /AC TO 7777, LINK=0
0373 1072 TAD K7777 /AC TO 7776, LINK TO 1
0374 7420 SNL
0375 7402 HLT /CARRY DID NOT PROPAGATE TO LINK
/MAKE AC=7777 FOR EASE OF CHECKING RESULT OF PREVIOUS TAD
0376 7001 IAC /AC TO 0000
0377 7040 CMA
0400 7440 SEA
0401 7402 HLT /CARRY DID NOT PROPAGATE CORRECTLY
/OR TAD USED INCORRECT OPERAND
/*****
/TEST ADDER'S ABILITY TO GENERATE CARRY WHEN ACN=1, MDN=1, AND CARRY IN N=0 IN ODD NUMBERED BIT POSITIONS
0402 1056 TAD K2525 /AC TO 2525, LINK = 1
0403 1056 TAD K2525 /AC TO 5252
0404 1056 TAD K2525 /AC TO 7777, LINK = 1 (NO CARRY GENERATED TO LINK)
0405 7040 CMA /AC TO 0000, LINK = 1
0406 7440 SEA
0407 7402 HLT /CARRY FAILED IN AN ODD BIT POSITION
0410 7420 SNL
0411 7402 HLT /LINK COMPLEMENTED WHEN NO CARRY OUT EXPECTED
/*****
/TEST ADDER'S ABILITY TO GENERATE CARRY WHEN ACN=1, MDN=1, AND CARRY IN N=0 IN ALL EVEN BIT POSITIONS
0412 1064 TAD K5252 /AC TO 5252, LINK=1
0413 1064 TAD K5252 /AC TO 2524, LINK TO 0
0414 1064 TAD K5252 /AC TO 7776, LINK=0
0415 7001 IAC /AC TO 7777, LINK=0
0416 7040 CMA /AC TO 0000, LINK=0
0417 7440 SEA
0420 7402 HLT /CARRY FAILED IN AN EVEN BIT POSITION
0421 7430 SEL
0422 7402 HLT /CARRY OUT FAILED TO COMPLEMENT LINK
/*****
/TEST AND INSTRUCTION TO NOT SET ANY AC BITS WHEN AC =0000
0423 7000 XOR06, NOP/JMS POINT FOR XOR
/*****
0424 0072 AND K7777 /AC=0000, LINK=0

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0425 7450 SNA
0426 7430 SZL
0427 7402 HLT
/***** /AND SET BIT WHEN AC INITIALLY CLEAR, OR AND SET LINK
/TEST AND INSTRUCTION TO CLEAR ALL AC BITS WHEN USING AN OPERAND OF 0000
0430 1072 TAO K7777 /AC TO 7777, LINK = 0
0431 0040 AND K0 /AC TO 0000, LINK = 0
0432 7450 SNA
0433 7430 SZL
0434 7402 HLT
/***** /AND FAILED TO CLEAR ALL AC BITS, OR SET LINK
/TEST AND INSTRUCTION TO NOT CLEAR ANY AC BITS WHEN AC=7777 AND M0=7777
0435 1072 TAO K7777 /AC TO 7777
0436 0072 AND K7777 /AC=7777, LINK=0
0437 7040 CMA /AC TO 0000, LINK = 0
0440 7450 SNA
0441 7430 SZL
0442 7402 HLT
/***** /AND OF 7777 CLEARED AC BIT OR SET LINK
/TEST FOR ADJACENT PIN SHORTS IN "AND" CIRCUITRY
0443 1056 TAO K2525 /AC TO 2525
0444 0064 AND K2525 /AC TO 0000
0445 7440 SNA
0446 7402 HLT
/***** /"AND" DID NOT CLEAR AC, POSSIBLE ADJACENT PIN SHORTS IN AND CIRCUITRY
XOR07, NOP/JMS POINT FOR XOR
/TEST ADDER=0 CIRCUITRY
/*****
/TEST BIT 11 INPUT TO ADDER=0
0450 1041 ADDZER, TAO K1 /AC TO 0001
0451 7450 SNA
0452 7402 HLT
/***** /ADDER=0 OPEN ON BIT 11 INPUT
/TEST BIT 10 INPUT TO ADDER=0
0453 7200 CLA /AC TO 0000
0454 1042 TAO K2 /AC TO 0002
0455 7450 SNA
0456 7402 HLT
/***** /ADDER=0 OPEN ON BIT 10 INPUT
/TEST BIT 9 INPUT TO ADDER=0
0457 7200 CLA /AC TO 0000
0460 1043 TAO K4 /AC TO 0004
0461 7450 SNA
0462 7402 HLT
/***** /ADDER=0 OPEN ON BIT 9 INPUT
/TEST BIT 8 INPUT TO ADDER=0
0463 7200 CLA /AC TO 0000
0464 1044 TAO K10 /AC TO 0000
0465 7450 SNA
0466 7402 HLT
/***** /ADDER=0 OPEN ON BIT 8 INPUT
/TEST BIT 7 INPUT TO ADDER=0
0467 7200 CLA /AC TO 0000
0470 1045 TAO K20 /AC TO 0020
0471 7450 SNA

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0472 7402 HLT
/***** /ADDER=0 OPEN ON BIT 7 INPUT
/TEST BIT 6 INPUT TO ADDER=0
0473 7200 CLA /AC TO 0000
0474 1046 TAO K40 /AC TO 0000
0475 7450 SNA
0476 7402 HLT
/***** /ADDER=0 OPEN ON BIT 6 INPUT
/TEST BIT 5 INPUT TO ADDER=0
0477 7200 CLA /AC TO 0000
0500 1050 TAO K100 /AC TO 0000
0501 7450 SNA
0502 7402 HLT
/***** /ADDER=0 OPEN ON BIT 5 INPUT
XOR08, NOP/JMS POINT FOR XOR
/TEST BIT 4 INPUT TO ADDER=0
0504 7200 CLA /AC TO 0000
0505 1051 TAO K200 /AC TO 0200
0506 7450 SNA
0507 7402 HLT
/***** /ADDER=0 OPEN ON BIT 4 INPUT
/TEST BIT 3 INPUT TO ADDER=0
0510 7200 CLA /AC TO 0000
0511 1052 TAO K400 /AC TO 0400
0512 7450 SNA
0513 7402 HLT
/***** /ADDER=0 OPEN ON BIT 3 INPUT
/TEST BIT 2 INPUT TO ADDER=0
0514 7200 CLA /AC TO 0000
0515 1131 TAO K1000 /AC TO 1000
0516 7450 SNA
0517 7402 HLT
/***** /ADDER=0 OPEN ON BIT 2 INPUT
/TEST BIT 1 INPUT TO ADDER=0
0520 7200 CLA /AC TO 0000
0521 1055 TAO K2000 /AC TO 0000
0522 7450 SNA
0523 7402 HLT
/***** /ADDER=0 OPEN ON BIT 1 INPUT
/TEST BIT 0 INPUT TO ADDER=0
0524 7200 CLA /AC TO 0000
0525 1061 TAO K4000 /AC TO 4000
0526 7450 SNA
0527 7402 HLT
/***** /ADDER=0 OPEN ON BIT 0 INPUT
XOR09, NOP/JMS POINT FOR XOR
/TEST RAR TO ROTATE AND NOT DROP ANY BITS
0531 7200 RARTST, CLA /AC TO 0000
0532 1056 TAO K2525 /AC TO 2525
0533 7100 CLL /LINK TO 1
0534 7020 CML /AC TO 5252, LINK=1
0535 7010 RAR
0536 7420 SNL
0537 7402 HLT
/***** /RAR DID NOT SHIFT AC11 TO LINK
0540 7040 CMA /AC TO 2525, LINK=1

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0541 0064 AND K5252 /AC=0000, LINK=1
0542 7440 SZA
0543 7402 HLT /RAR DROPPED BIT OR DID NOT SHIFT
/ANY BIT SET IN AC INDICATES POSITION OF DROPPED BIT
/*****
/TEST RAR TO NOT PICK UP ANY BITS
0544 1056 TAD K2525 /AC TO 2525, LINK=1
0545 7010 RAR /AC TO 5252, LINK=1
0546 0056 AND K2525 /AC TO 0000, LINK=1
0547 7450 SNA
0548 7420 SNL
0551 7402 HLT /RAR PICKED UP BIT, POSITION OF BIT PICK UP IS
/INDICATED BY BIT(S) SET IN AC
/*****
/TEST RAL TO SHIFT AND NOT DROP ANY BITS
0552 1064 TAD K5252 /AC TO 5252 LINK=1
0553 7004 RAL /AC TO 2525 LINK=1
0554 7420 SNL
0555 7402 HLT /RAL DROPPED LINK BIT
0556 7040 CMA /AC TO 5252 LINK=1
0557 0056 AND K2525 /AC TO 0000 LINK=1
0560 7440 SZA
0561 7402 HLT /RAL DROPPED BIT OR DID NOT SHIFT
/FAILING BIT POSITIONS ARE SET IN AC
/*****
/TEST RAL TO NOT PICK UP ANY BITS
0562 1064 TAD K5252 /AC TO 5252 LINK=1
0563 7004 RAL /AC TO 2525 LINK=1
0564 0064 AND K5252 /AC TO 0000 LINK=1
0565 7440 SZA
0566 7402 HLT /RAL PICKED UP BIT, BITS SET IN AC INDICATE FAILING POSITIONS
/*****
/TEST RTR TO SHIFT TWICE AND NOT DROP ANY BITS
0567 7100 CLL /CLEAR LINK
0570 1062 TAD K4444 /AC TO 4444 LINK=0
0571 7012 RTR /AC TO 1111 LINK=0
0572 7430 SZL
0573 7402 HLT /RTR PICKED UP LINK BIT
0574 7040 CMA /AC TO 6666
0575 0053 AND K1111 /AC TO 0000 LINK=0
0576 7440 SZA
0577 7402 HLT /RTR DROPPED BIT OR DID NOT SHIFT TWICE
/BIT SET IN AC INDICATES BIT DROPPED
0600 7000 XOR10, NOP/JMS POINT FOR XOR
/*****
/TEST RTR TO NOT PICK UP ANY BITS
0601 1062 RTRPCK, TAD K4444 /AC TO 4444
0602 7012 RTR /AC TO 1111
0603 0066 AND K5666 /AC TO 0000
0604 7440 SZA
0605 7402 HLT /RTR PICKED UP BIT, BIT SET IN AC INDICATES FAILING POSITION.
/*****
/TEST RTL TO SHIFT AND NOT DROP BITS
0606 1053 RTLDRP, TAD K1111
0607 7006 RTL /AC TO 4444 LINK=0

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0610 7430 SZL
0611 7402 HLT /RTL PICKED UP LINK BIT
0612 7040 CMA /AC TO 3333 LINK=0
0613 0062 AND K4444 /AC TO 0000 LINK=0
0614 7440 SZA
0615 7402 HLT /RTL DROPPED BIT OR DID NOT SHIFT
/BIT SET IN AC INDICATES BIT DROPPED
/*****
/TEST RTL TO NOT PICK UP ANY BITS
0616 1053 RTLPCCK, TAD K1111 /AC TO 1111
0617 7006 RTL /AC TO 4444
0620 0060 AND K3333 /AC TO 0000
0621 7440 SZA
0622 7402 HLT /RTL PICKED UP BIT, BIT SET IN AC INDICATES FAILING POSITION
/*****
/TEST BSW TO SWAP AND NOT DROP BITS ON CHANGE LINK
0623 7100 BSWDRP, CLL /INITIALIZE LINK TO ZERO
0624 1047 TAD K77 /AC TO 0077 LINK=0
0625 7002 BSW /AC TO 7700 LINK=0
0626 7430 SZL
0627 7402 HLT /BSW SET LINK
0630 1050 TAD K100 /AC TO 0000 LINK TO 1
0631 7440 SZA
0632 7402 HLT /BSW DID NOT SWAP OR PICKED UP BIT.
/*****
/TEST BSW FOR ADJACENT PIN SHORTS AND DROPPED BITS
0633 1067 BSWPCK, TAD K7700 /AC TO 7700 LINK=1
0634 7002 BSW /AC TO 0077 LINK=1
0635 7420 SNL
0636 7402 HLT /BSW CLEARED LINK
0637 7040 CMA /AC TO 7700
0640 1050 TAD K100 /AC TO 0000, LINK TO 0
0641 7440 SZA
0642 7402 HLT /BSW PICKED UP OR DROPPED BIT(S).
XOR11, NOP/JMS POINT FOR XOR
/*****
/TEST OF GROUP 3 OPERATES AND MQ REGISTER FOLLOWS
/*****
/TEST MQL TO CLEAR AC
0644 1072 MQLTST, TAD K7777 /AC TO 7777
0645 7421 MQL /AC TO MQ, 0 TO AC.
0646 7440 SZA
0647 7402 HLT /MQL DID NOT CLEAR AC
/*****
/TEST SWP TO EXCHANGE MQ AND AC, TEST RESULTS OF PREVIOUS MQL.
0650 7521 SWPTST, SWP /AC TO MQ, MQ TO AC, AC=7777 MQ=0000
0651 7040 CMA /AC TO 0000
0652 7440 SZA
0653 7402 HLT /SWP DID NOT LOAD AC WITH MQ, OR MQL DID NOT LOAD MQ, OR
/SWP OR MQL DROPPED A BIT.
0654 7040 CMA /AC TO 7777 MQ=0000
/*****
/TEST MQ REGISTER FOR OUTPUTS STUCK HIGH
0655 7521 SWP /AC TO 0000 MQ TO 7777
0656 7440 SZA

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0657 7402      HLT          /SWP PICKED UP BIT IN AC,
/*****
/TEST MQ FOR ADJACENT PIN SHORTS BY TESTING FOR DROPPED BITS
0660 1056      TAO          K2525 /AC TO 2525 MQ=7777
0661 7421      MQL          /AC TO 0000 MQ TO 2525
0662 7521      SWP          /AC TO 2525 MQ TO 0000
0663 7040      CMA          /AC TO 5252
0664 0056      AND          K2525 /AC TO 0000
0665 7440      SZA          HLT
0666 7402      HLT          /MQL OR SWP DROPPED BIT, POSSIBLE ADJACENT PIN SHORTS
                              /IN MQ, BIT SET IN AC INDICATES POSITION OF FAILURE
/*****
/TEST FOR ADJACENT PIN SHORTS IN MQ BY TESTING FOR BITS PICKED UP
XOR13, NOP/JMS POINT FOR XOR
0667 7000      TAO          K2525 /AC TO 2525 MQ=0000
0670 1056      MQL          /AC TO 0000 MQ TO 2525
0671 7421      SWP          /AC TO 2525 MQ TO 0000
0672 7521      AND          K5252 /AC TO 0000
0673 0064      SZA          HLT
0674 7440      HLT          /MQL OR SWP PICKED UP BIT, POSSIBLE ADJACENT PIN SHORT IN MQ,
                              /BIT SET IN AC INDICATES POSITION OF FAILURE,
0676 7100      CLL          CMA
0677 7040      CMA          /AC TO 7777
/*****
/TEST MQA TO OR THE MQ WITH THE AC, ENTER WITH AC=7777 MQ=0000 LINK=0
0700 7421      MQL          /AC TO 0000 MQ TO 7777
0701 7501      MQA          /AC TO 7777 MQ=7777
0702 7430      SZL          /MQA SET LINK
0703 7402      HLT          /AC TO 0000 MQ=7777
0704 7040      CMA          /MQA DID NOT OR MQ WITH AC
0705 7440      SZA          /AC TO 7777 MQ TO 0000
0706 7402      HLT          /MQA CLEARED MQ
0707 7521      SWP          /TEST CAM TO CLEAR AC AND MQ
0710 7450      SNA          CAMTST, SWP
0711 7402      HLT          /AC TO 0000 MQ TO 7777
                              /AC TO 7777 MQ=7777
0712 7521      SWP          CMA
0713 7040      CMA          /SET LINK
0714 7020      CML          /CLEAR AC AND MQ
0715 7621      CAM          /CAM DID NOT CLEAR AC
0716 7440      SZA          SNL
0717 7402      HLT          /CAM CLEARED LINK
0720 7420      SNL          /AC=0000 MQ=0000
0721 7402      HLT          /CAM DID NOT CLEAR MQ
0722 7521      SWP          /TEST ACL TO LOAD ZEROES FROM MQ TO AC
0723 7440      SZA          ACLTST, CMA
0724 7402      HLT          /AC TO 7777 MQ=0000
0725 7040      CMA          /AC TO 0000 MQ=0000
0726 7701      ACL          /ACL DID NOT LOAD 0000 TO AC
0727 7440      SZA          SNL
0730 7402      HLT
0731 7420      SNL

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0732 7402      HLT          /ACL CLEARED LINK
0733 7040      CMA          /AC TO 7777
0734 7521      SWP          /AC TO 0000 MQ TO 7777
0735 7440      SZA          HLT
0736 7402      HLT          /ACL CHANGED MQ
/*****
/TEST OF DCA AND ISZ, DIRECT ADDRESSING TO PAGE ZERO FOLLOWS
/*****
/TEST DCA TO STORE ALL 1'S, CLEAR AC, AND NOT AFFECT LINK
DCAZTS, CML
0737 7020      TAO          K7777 /SET LINK
0740 1072      DCA          TESLOC /AC TO 7777 LINK=0
0741 3112      SZA          /AC TO 0000 LINK=0
0742 7440      HLT          /DCA DID NOT CLEAR AC, OR DCA SKIPPED
0743 7402      SZL          /DCA SET LINK
0744 7430      HLT          /AC TO 7777
0745 7402      TAO          TESLOC
0746 1112      CMA          /DCA DID NOT STORE, OR DCA STORED TO WRONG ADDRESS
0747 7040      SZA          /TEST DCA TO STORE ALL ZEROES
0750 7440      HLT          XOR15, NOP/JMS POINT
0751 7402      HLT          DCA TESLOC /0000 TO LOCATION "TESLOC"
                              TAO TESLOC /0000 TO AC
0752 7000      DCA          /DCA DID NOT STORE ALL 0'S, BIT SET IN AC
0753 3112      TAO          /INDICATES FAILING BIT POSITION,
0754 1112      SZA          /TEST ISZ TO INCREMENT WITHOUT SKIPPING
0755 7440      HLT          ISZTS, ISZ TESLOC /LOCATION "TESLOC" TO 0001
0756 7402      HLT          /ISZ SKIPPED WHEN NO OVERFLOW, OR ISZ SET AC BIT
                              /ISZ SET LINK
0757 2112      TAO          TESLOC /AC TO 0001 LINK=0
0760 7440      TAO          K7777 /AC TO 0000, LINK TO 1
0761 7402      SZA          /ISZ DID NOT INCREMENT BY 1
0762 7430      HLT          /TEST ISZ TO SKIP ON OVERFLOW
0763 7402      HLT          TAO K7777 /AC TO 7777 LINK=1
0764 1112      DCA          TESLOC /LOCATION "TESLOC" TO 7777, AC TO 0000
0765 1072      ISZ          /SHOULD SKIP
0766 7440      HLT          /ISZ DID NOT SKIP ON OVERFLOW
0767 7402      HLT          /OVERFLOW ON ISZ SET LINK
                              /0000 TO AC
0770 1072      TAO          /ISZ DID NOT INCREMENT 7777 TO 0000, OR ISZ
0771 3112      DCA          /AFFECTED AC ON OVERFLOW
0772 2112      ISZ          /TEST OF "AND", TAO, ISZ, AND DCA, DIRECT ADDRESSING TO SAME PAGE
0773 7402      HLT          /*****
0774 7420      SNL          /TEST TAO TO ADDRESS SAME PAGE DIRECT
0775 7402      HLT
0776 1112      TAO          TESLOC
0777 7440      SZA
1000 7402      HLT

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1001 1203 TODSTS, TAD ,+2 /AC TO 2526 LINK=0
1002 7410 SKP
1003 2526 /OPERAND FOR TAD SAME PAGE TEST
1004 1064 TAD K5252 /AC TO 0000, LINK TO 1
1005 7440 SZA
1006 7402 HLT /TAD TO SAME PAGE DIRECT FAILED
/*****
/TEST DCA TO SAME PAGE DIRECT
DODSTS, TAD K7777 /AC TO 7777
1010 3212 DCA ,+2 /AC TO 0000
1011 7410 SKP
1012 0000 /TEST LOCATION FOR DCA TO SAME PAGE TEST
1013 1212 TAD ,+1 /AC TO 7777
1014 7040 CMA /AC TO 0000
1015 7440 SZA
1016 7402 HLT /DCA TO SAME PAGE FAILED
1017 3212 DCA ,+5 /CLEAR TEST LOCATION FOR POSSIBLE SECOND PASS
/*****
/TEST ISZ TO SAME PAGE DIRECT TO SKIP
XOR16, NOP/JMS POINT FOR XOR
1021 1072 ISDSTS, TAD K7777 /AC TO 7777
1022 3224 DCA ,+2 /AC TO 0000
1023 7410 SKP
1024 0000 /ISZ TEST LOCATION
1025 2224 ISZ ,+1 /SHOULD SKIP
1026 7402 HLT /ISZ DID NOT SKIP
/*****
/TEST ISZ TO SAME PAGE DIRECT TO NOT SKIP
1027 2224 ISZ ,+3 /SHOULD NOT SKIP
1030 7410 SKP
1031 7402 HLT /ISZ SKIPPED WHEN NO SKIP EXPECTED
/*****
/TEST "AND" TO SAME PAGE DIRECT TO CLEAR ALL AC BITS
1032 1064 ANDSTS, TAD K5252 /AC TO 2526
1033 0235 AND ,+2 /AC TO 0000
1034 7410 SKP
1035 2525 /OPERAND FOR AND SAME PAGE TEST
1036 7440 SZA
1037 7402 HLT /AND TO SAME PAGE DIRECT FAILED
/*****
/TESTS OF TAD, "AND", DCA AND ISZ INDIRECT THRU PAGE ZERO FOLLOW
/TEST OF TAD THRU PAGE ZERO INDIRECT
1040 1513 TDISTS, TAD I POINTR /AC TO 2526
1041 1064 TAD K5252 /AC TO 0000
1042 7440 SZA
1043 7402 HLT /TAD INDIRECT THRU PAGE ZERO FAILED, OR AUTO-INDEXED,
/*****
/TEST DCA INDIRECT THRU PAGE ZERO
1044 1072 DCISTS, TAD K7777 /AC TO 7777
1045 3515 DCA I POINTB /7777 TO LOCATION "POINTC", AC TO 0000
1046 1116 TAD POINTC /AC TO 7777
1047 7040 CMA /AC TO 0000
1050 7440 SZA
1051 7402 HLT /DCA INDIRECT THRU PAGE ZERO FAILED

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/*****
/TEST ISZ INDIRECT THRU PAGE ZERO
1052 2515 ISISTS, ISZ I POINTB /LOCATION "POINTC" TO 0000, SKIP
1053 7402 HLT /ISZ INDIRECT THRU PAGE ZERO FAILED TO SKIP
/*****
/TEST "AND" INDIRECT THRU PAGE ZERO
1054 1056 ANISTS, TAD K2525 /AC TO 2525
1055 3116 DCA POINTC /AC TO 0000
1056 7000 XOR17, NOP/JMS POINT
1057 1072 TAD K7777 /AC TO 7777
1060 0515 AND I POINTB /AC TO 2525
1061 1065 TAD K5253 /AC TO 0000
1062 7440 SZA
1063 7402 HLT /AND INDIRECT THRU PAGE ZERO FAILED
/*****
/TESTS OF TAD, DCA, ISZ, & "AND" INDIRECT THRU SAME PAGE FOLLOW
/TEST TAD INDIRECT THRU SAME PAGE
1064 1666 TDISTS, TAD I ,+2 /AC TO 2526
1065 7410 SKP
1066 0114 POINTR+1 /POINTER FOR TAD INDIRECT THRU SAME PAGE
1067 1064 TAD K5252 /AC TO 0000
1070 7440 SZA
1071 7402 HLT /TAD INDIRECT THRU SAME PAGE FAILED
/*****
/TEST DCA INDIRECT THRU SAME PAGE
1072 1072 DCISTS, TAD K7777 /AC TO 7777
1073 3675 DCA I ,+2 /AC TO 0000
1074 7410 SKP
1075 0116 POINTC /POINTER FOR DCA INDIRECT THRU SAME PAGE
1076 1116 TAD POINTC /AC TO 7777
1077 7040 CMA
1100 7440 SZA
1101 7402 HLT /DCA INDIRECT THRU SAME PAGE FAILED
/*****
/TEST ISZ INDIRECT THRU SAME PAGE
1102 2705 ISISTS, ISZ I ,+3 /LOCATION "POINTC" TO 0000, SKIP
1103 7402 HLT /ISZ INDIRECT THRU SAME PAGE FAILED TO SKIP
1104 7410 SKP
1105 0116 POINTC
/*****
/TEST "AND" INDIRECT THRU SAME PAGE
1106 1056 ANISTS, TAD K2525 /AC TO 2525
1107 3116 DCA POINTC /AC TO 0000
1110 1072 TAD K7777 /AC TO 7777
1111 0713 AND I ,+2 /AC TO 2525
1112 7410 SKP
1113 0116 POINTC /POINTER FOR "AND" INDIRECT THRU SAME PAGE
1114 1065 TAD K5253 /AC TO 0000
1115 7440 SZA
1116 7402 HLT /"AND" INDIRECT THRU SAME PAGE FAILED
1117 7000 XOR18, NOP/JMS POINT
/*****
/TESTS OF AUTO-INDEX FOLLOW
/*****

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1120 1056 /TEST AUTO-INDEX TO NOT INCREMENT WHEN NOT INDIRECTLY ADDRESSED.
1121 3010 AINITS, TAD K2525 /AC TO 2525
1122 1010 DCA 10 /ADDRESS 10 TO 2525
1123 1065 TAD 10 /AC TO 2525
1124 7440 TAD K5253 /AC TO 0000
1125 7402 SEA
HLT /AUTO-INDEX INCREMENTED WHEN NOT INDIRECTLY ADDRESSED
/*****
/TEST AUTO-INDEX TO INCREMENT WHEN INDIRECTLY ADDRESSED
1126 3116 AIIOTS, DCA POINTC /CLEAR LOCATION "POINTC"
1127 1056 TAD K2525 /AC TO 2525
1130 3117 DCA POINTD /LOCATION "POINTD" TO 2525
1131 1115 TAD POINTB
1132 3010 DCA 10 /SET LOCATION 10 TO THE ADDRESS OF "POINTC"
1133 1410 TAD I 10 /LOCATION "POINTD"'S CONTENTS TO AC, AC TO 2525
1134 1065 TAD K5253 /AC TO 0000
1135 7440 SEA
1136 7402 HLT /AUTO-INDEX FAILED TO INCREMENT
/*****
/TEST AUTO-INDEX DECODER FROM BITS 0 THRU 3 BY ADDRESSING ADDRESS 1010
/INDIRECTLY AND INSURING THAT AUTO-INDEXING DID NOT TAKE PLACE.
1137 7000 XOR20, NOP/JMS POINT
1140 1210 AIBIT0, TAD 1010 /GET INITIAL CONTENTS OF 1010
1141 3112 DCA TESLOC /SAVE FOR LATER COMPARISON
1142 0610 AND I 1010 /REFERENCE 1010 INDIRECTLY
1143 1210 TAD 1010 /GET CONTENTS OF 1010
1144 7041 CIA /NEGATE IT
1145 1112 TAD TESLOC /COMPARE TO INITIAL CONTENTS
1146 7440 SEA /FINAL=INITIAL?
1147 7402 HLT /NO, AUTO INDEX OCCURRED FOR LOCATION 1010
/*****
/TEST BIT 8 INPUT OF AUTO-INDEX DECODER BY ADDRESSING LOCATION 0007
/INDIRECTLY AND CHECKING THAT AUTO-INDEXING DID NOT OCCUR
1150 3007 AIBIT0, DCA 7 /CLEAR LOCATION 0007
1151 1407 TAD I 7 /REFERENCE LOCATION 0007 INDIRECTLY
1152 7200 CLA /CLEAR AC
1153 1007 TAD 7 /GET CONTENTS OF LOCATION 0007
1154 7440 SEA /DOES IT STILL CONTAIN 0007?
1155 7402 HLT /NO, LOCATION 0007 AUTO-INDEXED
/*****
/TEST AUTO-INDEX DECODER INPUT FROM BITS 4 THRU 7, BY REFERENCING
/ADDRESS 0030 INDIRECTLY AND CHECKING THAT AUTO-INDEXING DID NOT OCCUR
1156 1030 AIBIT4, TAD 30 /GET INITIAL CONTENTS OF LOC 0030
1157 3112 DCA TESLOC /SAVE IT FOR COMPARISON
1160 0430 AND I 30 /REFERENCE 30 INDIRECTLY
1161 1030 TAD 30 /GET FINAL CONTENTS OF LOC 30
1162 7040 CMA
1163 7001 IAC /COMPLEMENT IT FOR COMPARE
1164 1112 TAD TESLOC /COMPARE TO INITIAL CONTENTS
1165 7440 SEA /WAS LOC 30 AUTO-INDEXED?
1166 7402 HLT /YES, LOC 30 AUTO-INDEXED
1167 7000 XOR19, NOP/JMS POINT FOR XOR
/*****
/TESTS OF INTERNAL IOT INSTRUCTIONS FOLLOW

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/*****
/TEST GTF TO SAVE LINK, AND READ ZERO FOR THE GT FLAG
1170 7100 GTFTS1, CLL /CLEAR LINK
1171 7020 CML /LINK TO 1
1172 6004 GTF
1173 7500 SMA
1174 7402 HLT /GTF DID NOT SAVE A 1 FOR LINK
1175 7004 RAL
1176 7510 SPA
1177 7402 HLT /GTF READ A 1 FOR GT FLAG, NO GT FLAG EXISTS IN B/A
1200 7300 CLA CLL
1201 6214 RDP /GET DATA FIELD
1202 7012 RTR
1203 7010 RAR /MOVE DF TO AC 9-11
1204 6224 RIF /GET INSTRUCTION FIELD
1205 3106 DCA SAVFLD /SAVE IF AND DF FOR EXTENDED FIELD INTERRUPT PROCESSING
1206 7000 XOR20, NOP/JMS POINT
/*****
/TEST RTF TO RESTORE LINK
1207 1061 RTFTS1, TAD K4000 /AC TO 4000
1210 1106 TAD SAVFLD /GET CORRECT IF AND DF INFORMATION
1211 6005 RTF /RESTORE LINK TO 1
1212 5213 JMP ,*1 /ENABLE INTERRUPT FOR POWER FAIL
1213 7420 SNL
1214 7402 HLT /RTF DID NOT RESTORE LINK TO A 1
/*****
/TEST GTF TO SAVE A LINK OF 0, AND INT REQUEST AND INT ENABLE OF 0
1215 6002 GTFTS2, IOF
1216 7300 CLA CLL /CLEAR AC AND LINK
1217 6004 GTF /GET FLAGS, LINK TO AC0
1220 0366 AND K7600 /ELIMINATE SAVE FIELD REGISTER AND USER BIT
1221 7440 SEA /LINK, INT REQUEST, AND INT ENABLE ALL ZERO?
1222 7402 HLT /NO, GTF DID NOT SAVE CORRECTLY.
/IF BIT0=1 LINK WAS SAVED AS 1 INSTEAD OF 0
/IF BIT2=1 INT REQUEST WAS SAVED AS 1 INSTEAD OF 0
/IF BIT4=1 INT ENABLE WAS SAVED AS 1 INSTEAD OF 0
/*****
/TEST SKIP ON GT FLAG TO NOT SKIP
1223 6006 SGTST, SGT /SKIP ON GT FLAG (DOESN'T EXIST IN B/A)
1224 7410 SKP
1225 7402 HLT /SGT SKIPPED WHEN NO GT FLAG EXISTS
/*****
/TEST CAF TO CLEAR AC AND LINK
1226 6001 IOV
1227 7040 CMA /AC TO 7777
1230 7020 CML /AC=7777 LINK TO 1
1231 6007 CAF /CLEAR ALL FLAGS, CLEARS AC AND LINK, AND INT ENABLE
1232 7450 SNA
1233 7430 SZL
1234 7402 HLT /CAF FAILED TO CLEAR AC OR LINK
/*****
/TEST SKON TO NOT SKIP WHEN INTERRUPT ENABLE IS CLEAR
1235 6000 SKON /SHOULD NOT SKIP
1236 7410 SKP
1237 7402 HLT /SKON SKIPPED WHEN INT ENABLE CLEAR, OR CAF FAILED

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/TO CLEAR INT ENABLE
/*****
/TEST SKON TO SKIP ON INTERRUPT ON AND TURN INTERRUPT OFF
1240 6001 IONTS1, ION /INTERRUPT ON,
1241 7000 XOR21, NOP/JMS POINT FOR XOR
1242 6000 SKON /SKIP IF INTERRUPT ON, TURN INTERRUPT OFF
1243 7402 HLT /ION DID NOT ENABLE INTERRUPT, OR SKON FAILED TO SKIP
/*****
/TEST THAT SKON TURNED OFF INTERRUPT
1244 6000 SKONT2, SKON /SHOULD NOT SKIP
1245 7410 SKP
1246 7402 HLT /SKON DID NOT TURN OFF INT, OR SKON SKIPS WHEN INT OFF
/*****
/TEST IOF TO DISABLE INTERRUPTS
1247 6001 IOFTS1, ION /ENABLE INTERRUPTS
1250 7000 NOP /ALLOWS TIME FOR INTERRUPT DELAY TO SET
1251 6002 IOF /TURN OFF INTERRUPT
1252 6000 SKON
1253 7410 SKP
1254 7402 HLT /IOF DID NOT DISABLE INTERRUPT
/*****
/TEST PROPER OPERATION OF INT ENABLE,
1255 6001 INTEN1, ION /INTERRUPT ON
1256 6004 GTF /SHOULD GET A 1 FOR INT ENABLE, AND A 1 FOR INT DELAY
1257 0051 AND K200 /MASK OUT INT ENABLE BIT
1260 7450 SNA /SKIP IF INT ENABLE BIT SET
1261 7402 HLT /ION DID NOT SET INT ENABLE OR GTF DID NOT GET A 1 FOR INT ENABLE
/*****
/TEST RTF TO SET INT ENABLE AND TO CLEAR LINK
1262 6002 RTFTS2, IOF /CLEAR INT ENABLE
1263 7300 CLA CLL
1264 7020 CML /SET LINK
1265 1106 TAD SAVFLD /GET IF AND OF INFORMATION
1266 6005 RTF /RESTORE FLAGS, LINK TO 0, SET INTERRUPT ENABLE
1267 5270 JMP ,*1 /ENABLE INTERRUPT FOR POWER FAIL USE
1270 7430 SZL
1271 7402 HLT /RTF DID NOT CLEAR LINK
1272 6000 SKON /SKIP IF INTERRUPT ON
1273 7402 HLT /RTF FAILED TO ENABLE INTERRUPTS
/*****
/TEST SRQ TO NOT SKIP WHEN NO INTERRUPT REQUEST
1274 6001 SRQTS1, ION /INTERRUPT ON
1275 7300 CLA CLL /GIVE POWER FAIL A CHANCE TO INTERRUPT
1276 6003 SRQ /SKIP ON INTERRUPT REQUEST
1277 7410 SKP
1300 7402 HLT /SRQ SKIPPED WHEN NO INTERRUPT, OR ILLEGAL INTERRUPT
/*****
/TESTS OF JUMPS AND JMS'S FOLLOW
/*****
/TEST JUMP DIRECT
1301 7000 JMPTS1, NOP
1302 7300 CLA CLL /CLEAR AC AND LINK
1303 1135 TAD KSTOP
1304 3000 DCA 0 /PUT HLT IN LOC 0 IN CASE JUMP FAILS TO GATE MD TO PC

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1305 5307 JMP ,*2 /FIRST JUMP TESTED
1306 7402 HLT /JMP FAILED TO JUMP DIRECT
/*****
/TEST JMS DIRECT
1307 4311 JMSTS1, JMS ,*2
1310 7402 HLT /JMS FAILED TO JUMP
1311 0000 JMSLOC, 0 /JMS ENTRY POINT, PC STORAGE.
1312 1311 TAD ,*1 /GET STORED PC
1313 7040 CMA
1314 7001 IAC /COMPLEMENT IT
1315 1120 TAD
1316 7440 SEA
1317 7402 HLT /JMS DID NOT STORE PROPER PC
/*****
/TEST JUMP INDIRECT TO JUMP CORRECTLY
1320 5722 JMPTS2, JMP 1 ,*2
1321 7402 HLT /JMP INDIRECT FAILED TO JUMP
1322 1323 ,*1 /POINTER FOR JMP INDIRECT ABOVE
/*****
/TEST JMS INDIRECT TO JUMP AND STORE PC
1323 4725 JMSTS2, JMS 1 ,*2
1324 7402 HLT /JMS INDIRECT FAILED TO JUMP
1325 1326 ,*1
1326 0000 JMSLOC, 0 /JMS INDIRECT PC STORAGE
1327 1326 TAD ,*1
1330 7040 CMA
1331 7001 IAC /COMPLEMENT STORED PC
1332 1121 TAD POINTF /COMPARE IT TO EXPECTED VALUE
1333 7440 SEA /WAS IT EQUAL?
1334 7402 HLT /NO, JMS INDIRECT STORED INCORRECT PC.
1335 7000 XOR22, NOP/JMS POINT FOR XOR
/*****
/TESTS OF ROMS D AND F FOLLOW
/*****
/THE ROMS ARE TESTED BY EXECUTING EVERY INSTRUCTION THAT CAUSES A UNIQUE
/ADDRESS INPUT TO THESE ROMS, FOR EACH OF THE INSTRUCTIONS
/EXECUTED, A TOTAL OF 8 PATTERNS OF AC, HQ, AND LINK CONTENTS ARE USED,
/TO TEST FOR ERRORS IN THE EXECUTED INSTRUCTION A SIMULATION OF THE
/INSTRUCTION IS ALSO PERFORMED, AND THE RESULTS OF THE INSTRUCTION ARE
/COMPARED TO THE SIMULATION, BARRING INTERMITTENT ERRORS, ALL FAILURES
/WILL BE IN THE EXECUTED INSTRUCTION, AND WILL BE DUE TO A ROM D OR F FAILURE.
/*****
/CREATE THE INSTRUCTION TO BE USED, INSTRUCTION RANGE IS 7XXB, 7XX1.
/*****
1336 3116 ROMTS1, DCA POINTC /INITIALIZE INSTRUCTION MAKER
1337 7410 SKP
1340 7300 NXTONE, CLA CLL
1341 2116 ISE POINTC
1342 7020 CML
1343 7000 XOR12, NOP/JMS POINT
1344 1116 TAD POINTC
1345 1067 TAD K7700
1346 7500 SNA /ALL COMBINATIONS TRIED?
1347 5524 JMP I CKSWIT /YES, TEST OVER, CHECK FOR I/O SIMULATOR.
1350 7300 CLA CLL

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1507 2112 /GROUP 2 SIMULATION
1510 5334 GR2GR3, ISE BIT11 /GROUP 2?
1511 2006 JMP: GROUP3 /NO, GROUP 3,
1512 5340 GROUP2, ISE BIT8 /REVERSE SENSE SKIPS?
1513 2005 JMP: REVSER /YES, GO DO REVERSE
1514 7500 ISE BIT5 /IS IT SMA?
1515 7410 SMA /YES
1516 2105 SKP: SOMSKP /SMA SKIPPED,
1517 2117 ISE BIT6 /IS IT SEA?
1520 7440 SEA /YES
1521 7410 SKP: SOMSKP /SEA SKIPPED
1522 2105 ISE BIT7 /IS IT SNL?
1523 2007 ISE /YES
1524 7420 SNL: SOMSKP /SAVE AC
1525 7410 SKP: SOMSKP /ANY SKIP?
1526 2105 ISE SKPPED /NO
1527 3007 OUT, DCA BIT7
1530 1105 TAD SOMSKP
1531 7450 SNA /REPLACE AC
1532 2104 ISE SKPPED /WAS IT CLA?
1533 7200 CLA /YES
1534 1007 TAD BIT7
1535 2004 ISE BIT4
1536 7200 CLA
1537 5523 JMP I TEST

/*****
/REVERSE SENSE SKIPPING FOR GROUP 2
1540 2005 REVSER, ISE BIT5 /WAS IT SPA?
1541 7500 SMA /YES,
1542 7410 SKP: SPA /SPA WOULD HAVE SKIPPED,
1543 5327 JMP OUT /SPA WOULD NOT HAVE SKIPPED
1544 2117 ISE BIT6 /WAS IT SNA?
1545 7440 SEA /YES,
1546 7410 SKP: SNA /SNA WOULD HAVE SKIPPED
1547 5327 JMP OUT /SNA WOULD NOT HAVE SKIPPED
1550 2007 ISE BIT7 /SZL?
1551 7420 SNL: /YES
1552 5326 JMP OUT-1 /QUALIFIED SKIP
1553 5327 JMP OUT /SZL WOULD NOT HAVE SKIPPED.
*****/

/*****
/GROUP 3 OPERATE SIMULATION
1554 2104 GROUP3, ISE SKPPED /INDICATE NO SKIP
1555 2004 ISE BIT4 /CLA?
1556 7200 CLA /YES
1557 2005 ISE BIT5 /IS IT MQA?
1560 5364 JMP: +4 /YES
1561 2007 ISE BIT7 /IS IT MQL?
1562 7421 MQL: /YES
1563 5523 JMP I TEST /COMPARE RESULTS
1564 2007 ISE BIT7 /IT IS MQA, IS IT ALSO MQL? (SWP)
1565 5370 JMP DOSWAP /YES, DO A SWP.
1566 7501 MQA /NO, JUST DO MQA.
1567 5523 JMP I TEST /COMPARE RESULTS
1570 3003 DOSWAP, DCA BIT3

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1571 7501 MQA /MQ TO AC
1572 3004 DCA BIT4 /SAVE PREVIOUS MQ
1573 1003 TAD BIT3 /GET PREVIOUS AC
1574 7421 MQL: /PUT IN MQ
1575 1004 TAD BIT4 /PUT PREVIOUS MQ IN AC,
1576 5523 JMP I TEST /TEST RESULTS

/*****
/COMPARE RESULTS OF SIMULATION TO ACTUAL RESULTS
*****/

/TEST LINKS
1577 3112 TESTS, DCA BIT11 /SAVE SIMULATED AC
1600 7004 RAL: /LINK TO AC 11
1601 3007 DCA BIT7 /SAVE SIMULATED LINK
1602 1007 TAD BIT7 /GET EXPECTED LINK
1603 1103 TAD LKWS /ADD IN LINK OBTAINED
1604 3003 DCA BIT3 /SET ERROR INDICATOR IF LINKS DIFFERENT
/RESET ERROR INDICATOR IF LINKS SAME

/*****
/LINKS AGREE, TEST THE AC CONTENTS
1605 1112 TAD BIT11 /GET EXPECTED AC CONTENTS
1606 7040 CMA /COMPLEMENT IT
1607 7001 IAC /GET ACTUAL RESULTS
1610 1101 TAD ACWAS /SAME?
1611 7440 SEA /NO, SET ERROR INDICATOR
1612 3003 DCA BIT3

/*****
/AC CONTENTS OK, TEST MQ CONTENTS
1613 7521 SWP /MQ TO AC
1614 3006 DCA BIT8 /SAVE MQ
1615 1006 TAD BIT8
1616 7040 CMA /COMPLEMENT IT
1617 7001 IAC /COMPARE TO ACTUAL RESULTS
1620 1102 TAD MQWAS /SAME?
1621 7440 SEA /NO, SET ERROR INDICATOR
1622 3003 DCA BIT3

/*****
/CHECK FOR SIMULATED INSTRUCTION AND ACTUAL INSTRUCTION TO BOTH HAVE
/SHIPPED OR BOTH TO HAVE NOT SKIPPED
1623 1104 SKPCHK, TAD SKPPED /GET SKIP INDICATOR
1624 7010 RAR: /AC 11 TO LINK
1625 7200 CLA /BOTH SKIP OR BOTH NOT SKIP?
1626 7420 SNL: /YES, BOTH SKIPPED OR BOTH DIDN'T
1627 5243 JMP: SIMERR /FAILING INSTRUCTION TO AC.
1630 1922 TAD I INSTR

//////////
1631 7402 HLT /THE INSTRUCTION IN THE AC SKIPPED WHEN EXECUTED, AND A
/SIMULATION OF THE SAME INSTRUCTION DID NOT, OR VICE VERSA.
/BARRING INTERMITTENT FAILURES, ROM D OR F IS FAULTY.
/DEPRESS CONTINUE FOR STATE OF AC, LINK, AND MQ WHEN THE
/INSTRUCTION WAS EXECUTED.

//////////
/GET DATA THAT WAS IN AC, MQ, AND LINK WHEN SKIP PROBLEM OCCURRED,
1632 7621 CAM:
1633 1110 TAD MQDATA /GET MQ DATA

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1634 7521 SWP /PUT IT IN MQ
1635 1111 TAD LKDATA /GET LINK DATA
1636 7010 RAR /LOAD LINK
1637 7200 CLA
1640 1107 TAD ACDATA /LOAD AC DATA
1641 7402 HLT /AC, LINK, AND MQ ARE AS THEY WERE WHEN INSTRUCTION WAS EXECUTED,
/DEPRESS CONTINUE TO EXECUTE FAILING INSTRUCTION OVER.
1642 5522 JMP I INSTTR /GO EXECUTE INSTRUCTION AGAIN
/*****
/TEST FOR ANY SIMULATION DATA ERRORS
1643 1003 SIMERR, TAD BITS /GET ERROR INDICATOR
1644 7450 SNA /ANY SIMULATION ERRORS?
1645 5271 JMP TESTPT /NO
/*****
/DISPLAY THE ERROR INFORMATION
1646 7200 DISERR, CLA /ERRORS EXIST
1647 1522 TAD I INSTTR /GET FAILING INSTRUCTION
1650 7402 HLT /OPERATE INSTRUCTION FAILED. FAILING INSTRUCTION
/IS IN THE AC. DEPRESS CONTINUE FOR EXPECTED DATA,
/DEPRESS INTERMITTENTS, ROM D OR F AT FAULT.
/*****
/GET EXPECTED DATA FOR AC, MQ, AND LINK
1651 7300 GETEXP, CLA CLL
1652 1007 TAD BIT7 /GET EXPECTED LINK
1653 7010 RAR
1654 1006 TAD BIT8
1655 7421 MQL /LOAD EXPECTED MQ
1656 1112 TAD BIT11 /LOAD EXPECTED AC
1657 7402 HLT /AC, MQ, AND LINK CONTAIN EXPECTED DATA
/DEPRESS CONTINUE FOR DATA FOUND
/*****
/GET DATA FOUND
1660 7300 GETFND, CLA CLL
1661 1103 TAD LKWS
1662 7010 RAR
1663 7200 CLA
1664 1102 TAD MQWS
1665 7421 MQL /LOAD MQ AS IT WAS FOUND
1666 1101 TAD ACWS /LOAD AC AS IT WAS FOUND
1667 7402 HLT /AC, MQ, AND LINK ARE AS THEY WERE FOUND
/DEPRESS CONTINUE TO EXECUTE SAME INSTRUCTION OVER
/*****
/DO SAME INSTRUCTION OVER FOR ERROR
1670 5742 DOAGAN, JMP I NXTPTT /JMP TO NXTPTT
/*****
/SIMULATION AGREES WITH ACTUAL, SEE IF ALL DATA PATTERNS HAVE
/BEEN TRIED WITH THIS INSTRUCTION.
1671 2177 TESTPT, ISZ DATPAT /ALL PATTERNS TRIED?
1672 5314 JMP TRNXT /NO, TRY NEXT PATTERN

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1673 3107 DCA ACDATA
1674 7000 XOR14, NOP/JMS POINT
1675 3110 DCA MQDATA
1676 3111 DCA LKDATA
1677 1522 TAD I INSTTR /GET INSTRUCTION
1700 7010 RAR /BIT11 TO LINK
1701 7430 SEL /HAS INSTRUCTION BEEN TRIED WITH BIT11=1?
1702 5743 JMP I NXTONN /YES, DO NEXT INSTRUCTION (JMP TO NXTONE)
1703 2522 ISZ I INSTTR /UPDATE INSTRUCTION
1704 0051 AND K200 /MASK OUT MQ TYPE BIT
1705 7450 SNA /MQ TYPE?
1706 5744 JMP I TESAGG /NO, GO DO IT
1707 7200 CLA
1710 1522 TAD I INSTTR
1711 0070 AND K7721 /MASK OUT BITS NOT ALLOWED(EAE)
1712 3522 DCA I INSTTR
1713 5744 JMP I TESAGG /JMP TO TESAGN
/*****
/CREATE NEXT DATA SET UP
1714 7000 TRNXT, NOP/JMS POINT
1715 1177 TAD DATPAT
1716 7010 RAR /AC BIT TO LINK
1717 7200 CLA
1720 1056 TAD K2525
1721 7420 SNL /AC TO BE SET?
1722 7200 CLA /NO
1723 3107 DCA ACDATA
1724 1177 TAD DATPAT
1725 7012 RTR
1726 7200 CLA
1727 1522 TAD I INSTTR
1730 7420 SNL /MQ TO BE SET?
1731 7200 CLA /NO
1732 3110 DCA MQDATA
1733 1177 TAD DATPAT
1734 7012 RTR
1735 7010 RAR
1736 7420 SNL /LINK TO BE SET?
1737 7200 CLA /NO
1740 3111 DCA LKDATA
1741 5742 JMP I NXTPTT /JMP TO NXTPAT
1742 1360 NXTPTT, NXTPAT
1743 1340 NXTONN, NXTONE
1744 1356 TESAGG, TESAGN
/*****
/TEST FOR I/O SIMULATOR
1745 7000 XOR29, NOP/JMS POINT
1746 1021 TAD 21 /GET HARDWARE DESIGNATOR
1747 0052 AND K400 /MASK OUT SIMULATOR BIT,
1750 7440 SNA /DO WE HAVE A SIMULATOR?
1751 5772 JMP I TSTCHX/CLA CLL FOR 1K /YES, DO PART 2
/TEST FOR HALT AT END OF PASS
1752 1021 LOPBAK, TAD 21 /NO
1753 7700 SMA CLA /FRONT PANEL AVAILABLE?

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1754 5357 JMP ,+3 /NO
1755 7404 QSR /YES, "OR" THE SWITCH REGISTER WITH AC
1756 7410 SKP
1757 1020 TAD 20 /NO, FRONT PANEL, USE 20 FOR SR
1760 0052 AND K400
1761 7640 SZA CLA /HALT ON PASS COMPLETE?
1762 7402 7402/HLT /YES, HALT,
1763 1021 TAD 21 /GET HARDWARE DESIGNATOR
1764 0050 AND K100
1765 7440 SZA /RUNNING ON XOR?
1766 5773 JMP I XORPNT/NOP FOR 1K /YES
1767 7000 XOR30, NOP/JMS POINT
1770 5771 JMP I STOVER /START OVER BUT SKIP INITIAL HALT
1771 0222 STOVER, TSQACL
1772 2200 TSTCHX, 2200
1773 4000 XORPNT, XORSTT
1777 1777 *1777
1777 0000 /USED FOR DATA BREAK
2200 2200

```

 /8A INSTRUCTION TEST, PART 2, TO BE RUN ONLY WITH THE 8A I/O SIMULATOR
 /MODULE, THIS SECTION COMPLETES TESTING OF THE 8A CPU MODULE.

 /NOTE: DURING ALL I/O TRANSFERS WITH THE 8A I/O SIMULATOR, THE "C"
 /LINES ARE ASSERTED FROM TP3 TIME OF THE IOT INSTRUCTION UNTIL TP3 TIME
 /OF THE FOLLOWING INSTRUCTION, IN ORDER TO TEST THE ADDRESSING OF ROM E,
 /I.E. "C" LINES ASSERTED WITH NO I/O PAUSE, THIS CONDITION SHOULD CAUSE
 /NO ENABLES TO BE OUTPUTTED FROM ROM E) AN IMPROPERLY BLASTED ROM E COULD
 /CAUSE ANY OF THE ERROR HALTS IN THE SUCCEEDING SECTION, OR IN THE ROM E
 /TESTS THAT FOLLOW LATER IN THE PROGRAM.

```

2200 7000 /CHECK FOR 8/A OR 8/E TYPE CPU
2201 7200 TSTF8E, NOP/JMS POINT
2202 6001 CLA
2203 1021 ION
2204 0046 TAD 21 /GET HARDWARE DESIGNATOR
2205 7650 AND K40 /MASK OUT 8/E BIT
2206 1353 SNA CLA /8/A?
2207 1034 TAD K410 /YES, MAKE A SKIP
2210 3777 TAD K7000 /NO, 8/E, MAKE A NOP
DCA TSDPNS /THIS ACTION PREVENTS TESTING "DEPOSIT
/ON-STOP" ON PDP-8/E'S

2211 1021 TAD 21
2212 0046 AND K40
2213 7640 SZA CLA /8/E?
2214 1034 TAD K7000 /YES
2215 3776 DCA DF8E8A /THIS ACTION MAKES UP FOR A SMALL TIMING
/DIFFERENCE IN THE HANDLING OF "USER MODE"
/BETWEEN 8/E'S AND 8/A'S.

/*****
/CHANGE POWER FAIL TO RETURN TO BEGINNING OF PART TWO IN THE EVENT
/OF A POWER FAILURE OCCURRING DURING PART TWO
2216 7240 CLA CMA
2217 3100 DCA PARTWO

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/*****
/SET UP INTERRUPT SKIP CHAIN IF ON ACT-8/E LINE
2220 1022 ACTINT, TAD 22 /GET HARDWARE WORD 2
2221 7004 RAL /ACT-8/E BIT TO LINK
2222 7700 SNA CLA /ON ACT-8/E LINE?
2223 5236 JMP XOR27 /NO, SKIP THE RESTORE
2224 1375 TAD 6244 /YES, GET A RMP INSTRUCTION FOR SKIP CHAIN.
2225 3001 DCA 1
2226 1374 TAD (JMP SKPCHN
2227 3002 DCA 2

/*****
/TEST THAT A JMS IS NOT DECODED AS AN IOT, CAUSING "I/O PAUSE".
2230 1354 JMSJS, TAD KJMPI4
2231 3005 DCA 5 /SET UP A RETURN FROM LOCATION 4
2232 7120 CLL CML /SET LINK
2233 4004 JMS 4 /THIS WILL CAUSE A RTF OPERATION
/IF "I/O PAUSE" IS ERRORNEOUSLY GENERATED.

/RETURN HERE IMMEDIATELY FROM LOCATION 4
2234 7440 SZA /AC STILL CLEAR?
2235 7402 HLT /NO, JMS TO LOCATION 4 SET THE AC,

/*****
/TEST FOR AC OUTPUTTED WHEN ALL C LINES=H (ROM E, ADDRESS 27)
2236 7000 XOR27, NOP/JMS POINT
2237 6145 TSTCLB, CLRAL
2240 7340 CLL CLA CMA /AC TO 7777
2241 6142 STROB /IOT WITH C0,C1,C2=H
2242 7650 SNA CLA /ENSURE AC IS STILL SET
2243 7402 HLT /AC WAS CLEARED WHEN C0,C1,C2=H
2244 6141 READA /READ BACK THE OUTPUT DATA, AC TO 7777
2245 7040 CMA /AC TO 0000, DROPPED BITS NOW APPEAR AS 1'S
2246 7440 SZA /ANY BITS DROP?

/A HALT AT THE FOLLOWING LOCATION CAN BE CAUSED BY SETTING THE I/O SIMULATOR
/BIT IN LOCATION 21, WHEN NO I/O SIMULATOR IS ON THE OMNIBUS, SEE SECTION
/4.2.1 FOR PROPER INITIALIZATION PROCEDURE
2247 7402 HLT /YES, BITS NOT OUTPUTTED/INPUTTED ARE
/NOW SET IN THE AC.

/*****
/TEST FOR OUTPUT AC AND CLEAR AC WHEN C0=L, C1=H, & C2=H (ROM E ADD 26)
2250 7000 XOR32, NOP/JMS POINT
2251 6145 TSTCLB, CLRAL /CLEAR SIMULATOR
2252 7240 CLA CMA
2253 6146 LOADA
2254 7440 SZA /WAS AC CLEARED?
2255 7402 HLT /NO, C0=L C1&C2=H DID NOT CLEAR AC
2256 6141 READA /GET OUTPUTTED DATA
2257 7040 CMA /AC TO 0000, ANY BITS DROPPED ARE NOW SET
2260 7440 SZA /ANY BITS DROP?
2261 7402 HLT /YES, BITS NOT INPUTTED/OUTPUTTED IN AC

/*****
/TEST FOR "OR" INPUT TO AC WHEN C0&C2=H AND C1=L (ROM E ADDRESS 25)
2262 7000 XOR46, NOP/JMS POINT
2263 6145 TSTCLB, CLRAL
2264 1055 TAD K2000
2265 6147 LOADB /C1=LQH

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2266 1056      TAO      K2525
2267 6146      LOADA
2270 1064      TAO      K5252      /REGISTER A TO 2525
2271 6142      STROB      /AC TO 2525
2272 7040      CMA      /AC TO 7777
2273 7440      SZA      /AC TO 0000, BITS NOT "OR'ED" SET IN AC
2274 7402      HLT      /ANY BITS NOT OR'ED?
                        /YES, BITS SET IN AC DID NOT "OR" INPUT
/*****
/TEST FOR JAM INPUT WHEN C0&C1=L AND C2=H (ROM E ADDRESS 24)
XOR47, NOP/JMS POINT
TSTCLD, CLRAL      /CLEAR SIMULATOR
                        /AC TO 2525
2275 7000      TAO      K2525
2276 6145      LOADA      K6000
2277 1056      TAO      K5252      /C0&C1=LOW
2300 6146      STROB      K5252      /AC TO 2525
2301 1355      CMA      K5253      /AC TO 0000
2302 6147      SZA      /JAM INPUT DONE CORRECTLY?
2303 1064      HLT      /YES,
2304 6142      /NO, JAM INPUT FAILED,
2305 1065
2306 7450
2307 5312
2310 7402
/*****
2311 7402      IOFHLT, HLT      /IOF FAILED TO DISABLE INTERRUPT, SEE BELOW.
/TEST REMAINDER OF ROM J AND ASSOCIATED CIRCUITRY
XOR31, NOP/JMS POINT
2312 7000      CLRAL      /CLEAR SIMULATOR
2313 6145      TAO      (IOFHLT      /SET UP INTERRUPT RETURN
2314 1373      DCA      7      /TO IOFHLT
2315 3007      TAO      (JMP I 7      /SET UP SKIP CHAIN FOR RETURNS
2316 1372      IOF      /TURN OFF INTERRUPT
2317 6002      DCA      CHNCON
2320 3036      TAO      K200      /CAUSE AN INTERRUPT
2321 1051      LOADA
2322 6146      CNTENA
2323 6140
/*****
/TEST SRQ TO SKIP ON INTERRUPT REQUEST
2324 6003      TSTSRQ, SRQ      /SKIP ON INTERRUPT REQUEST
2325 7402      HLT      /SRQ DID NOT SKIP, OR SIMULATOR DID NOT
                        /CAUSE AN INTERRUPT
/*****
/TEST GTF TO SAVE INTERRUPT LINE
2326 6004      GTFSS, GTF      /GET FLAGS
2327 0131      AND      K1000      /MASK INTERRUPT REQUEST
2330 7650      SNA CLA
2331 7402      HLT      /GTF DID NOT SAVE INTERRUPT LINE
2332 7000      XOR33, NOP/JMS POINT
/*****
/TEST FOR INTERRUPT TO OCCUR ONLY WHEN "FSET LOW"
TSFSET, CLRAL
2333 6145      IOV
2334 6001      TAO      (FSETER
2335 1371      DCA      7
2336 3007      TAO      K200
2337 1051

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2340 6146      LOADA      /CAUSE AN INTERRUPT
2341 6140      CNTENA
2342 1513      TAO I POINTR      /INTERRUPT AFTER EXECUTE, AC TO 2526
2343 7402      HLT      /WE SHOULD HAVE INTERRUPTED
2344 1343      FSETER, TAO      ,=1
2345 3036      DCA      CHNCON
2346 1074      TAO      SAVAC      /GET INTERRUPTED AC
2347 1064      TAO      K5252
2350 7440      SZA
2351 7402      HLT
                        /TAO INDIRECT DID NOT WORK CORRECTLY WHEN
                        /INTERRUPT WAS UP DURING DEFER AND
                        /EXECUTE, MAY HAVE INTERRUPTED AT END OF
                        /FETCH OR DEFER
2352 5770      JMP      XOR34      /CONTINUE ON NEXT PAGE
2353 0410      K410, 0410
2354 5404      KJMP14, JMP I 4
2355 6000      K6000, 6000
2370 2400
2371 2344
2372 5407
2373 2311
2374 5023
2375 6244
2376 2451
2377 3350
2400      PAGE
/*****
/TEST OPERATION OF INTERRUPT DELAY
XOR34, NOP/JMS POINT
2401 1377      TSINTO, TAO      (DLYHLT
2402 3007      DCA      7      /SET UP INTERRUPT RETURN
2403 1376      TAO      (JMP I 7
2404 3036      DCA      CHNCON
2405 1051      TAO      K200
2406 6146      LOADA      /CAUSE INTERRUPT PULSE
2407 6140      CNTENA
2410 6001      IOV
2411 6002      IOF
2412 7410      SKP
2413 7402      DLYHLT, HLT      /INTERRUPT DELAY NOT WORKING
/*****
/TEST OPERATION OF INTERRUPT
XOR35, NOP/JMS POINT
2414 7000      TSTINT, CLRAL
2415 6145      IOV
2416 6001      TAO      (TSTPCI
2417 1375      DCA      7      /SET UP INTERRUPT RETURN
2420 3007      TAO      K200
2421 1051      LOADA      /CAUSE INTERRUPT
2422 6146      CNTENA
2423 6140
2424 0101      K101, AND      101      /SHOULD INTERRUPT AT THE END OF THIS INSTRUCTION
2425 7402      INTADD, HLT      /INTERRUPT FAILED
/*****

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2426 1225 /TEST THAT INTERRUPT STORED CORRECT PC
2427 3036 TSTPCI, TAD .-1
2430 6001 DCA CHNCON /SET UP FOR UNEXPECTED INTERRUPT
2431 1374 ION
2432 1000 TAD (-INTADD
2433 7440 TAD 0
2434 7402 SEA
HLT /INTERRUPT DID NOT STORE CORRECT PC
/*****
/TEST THAT USER MODE DISABLES A HLT INSTRUCTION
2435 7000 XOR36, NOP/JMS POINT
2436 6145 CLRAL
2437 1050 TAD K100
2440 6146 LOADA /CAUSE A PULSE ON USER MODE
2441 6140 CNTENA
2442 7602 7602/HLT CLA /ONLY HALTS IF USER MODE FAILS TO DISABLE
/*****
/TEST THAT USER MODE DISABLES AN IOT INSTRUCTION
/THIS TEST DIFFERS SLIGHTLY IN TIMING BETWEEN 8/A AND 8/E TYPE PROCESSORS
2443 7000 XOR37, NOP/JMS POINT
2444 6145 CLRAL
2445 1050 TAD K100 /USER MODE PULSE BIT
2446 7001 IAC /3 CYCLE DELAY
2447 6146 LOADA
2450 6140 CNTENA
2451 0000 DFB8A, AND 0/NOP
/((NOP FOR PDP-8/E), THE INSTRUCTION DIFFERENCES EXIST
/FOR THE FOLLOWING REASON: THE SIMULATOR PULLS THE USER MODE LINE
/ON THE THIRD TP1 AFTER THE CNTENA INSTRUCTION. THE USER MODE
/LINE WILL REMAIN LOW UNTIL THE FOLLOWING TP1 PULSE. FOR PDP-8/A'S
/THE STATE OF THE USER MODE LINE IS GATED TO A F/F AT TP1 TIME
/AND THE F/F IS USED TO DISABLE I/O PAUSE AT TP3 TIME. PDP-8/E'S
/USE THE USER MODE DIRECTLY AT TP3 TIME TO DISABLE I/O PAUSE.
/THEREFORE IN THE PROGRAM, ALTHOUGH THE USER MODE LINE GOES LOW
/AT TP1 TIME OF THE NOP AT DFB8A+1, IT DOES NOT ARRIVE AT THE
/8/A CPU IN TIME TO SET THE F/F WHICH CHECKS THE USER MODE LINE
/AT TP1 TIME, SO I/O PAUSE IS NOT DISABLED DURING THE NOP.
/AT THE NEXT TP1 TIME, USER MODE REMAINS LOW LONG ENOUGH TO
/SET THE F/F, THEREFORE I/O PAUSE IS DISABLED FOR THE READA INSTRUCTION.
/FOR PDP-8/E'S, THE "AND 0"(USES FETCH & EXECUTE), IS CHANGED
/TO A "NOP"(USES FETCH ONLY), THIS ELIMINATES ONE MAJOR STATE BETWEEN
/THE "CNTENA" AND THE "READA", SINCE IN EFFECT THE PDP-8/E SEES
/THE USER MODE LINE ONE MAJOR STATE EARLIER THAN THE 8/A IN THIS
/PARTICULAR INSTANCE.

2452 7000 NOP
2453 6141 READA /WON'T READ IF USER MODE DISABLES I/O PAUSE
2454 7640 SEA CLA

/THE FOLLOWING HALT CAN BE CAUSED BY SETTING BIT 6 IN LOCATION 21 TO
/SPECIFY AN 8/E TYPE CPU WHEN THE TEST IS ACTUALLY RUNNING ON A PDP-8/A.
/OR VICE VERSA,
2455 7402 HLT /USER MODE DID NOT DISABLE I/O PAUSE

/*****
/TEST SKIP LINE TO CPU BOARD AND OVERFLOW LINE FROM CPU BOARD, AND "LINK LOAD" LINE

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2456 7000 XOR38, NOP/JMS POINT
2457 7360 CLA CMA CLL CML /AC TO 7777 LINK TO 1
2460 3112 DCA TESLOC /SET UP FOR ISE SKIP
2461 6145 CLRAL /CLEAR SIMULATOR
2462 2112 ISE TESLOC /ISE SKIPS AND PUTS OVERFLOW LOW ON BUS
2463 7402 HLT /ISE DID NOT SKIP
2464 6144 SKPDFV /SKIP ON OVERFLOW
2465 7402 HLT /SKIP LINE FROM OMNIBUS STUCK HIGH
/OR OVERFLOW LINE TO BUS STUCK HIGH
2466 7430 SEL /DID LINK LOAD FROM "CLRAL" INSTRUCTION CLEAR LINK?
2467 7402 HLT /NO, LINK LOAD FAILED TO CLEAR LINK
/*****
/TEST ROM 4 ADDRESS 04, BY EXECUTING AN IOT WITH MD3 SET, CHECK TO INSURE
/OP1, OP2, OR OP3 NOT BEING DECODED, AND THAT NEITHER EXECUTE NOR DEFER
/IS BEING DONE,
2470 7000 XOR48, NOP/JMS POINT
2471 7300 ROMH04, CLA CLL
2472 1224 TAD K101 /PULSE TO USER MODE AFTER 3 CYCLE DELAY
2473 6146 LOADA
2474 6140 CNTENA
2475 7240 CLA CMA
2476 6770 6770
2477 7402 7402/HLT
/AC TO 7777
/IOT WITH BIT3 SET, ASSUMED NO DEVICE USES THIS IOT
/IF IOT WORKS CORRECTLY, USER PULSE WILL
/ARRIVE DURING THIS HLT AND PREVENT MACHINE FROM
/STOPPING. IF HALT OCCURS, ASSUMED THAT ROM H
/ADDRESS 04 IS TRANSLATING FOR A DEFER OR
/EXECUTE FOR IOT'S WITH BIT3 SET.(OR SIMULATOR
/MAY NOT BE PULSING USER MODE LINE)
/AC STILL SET?
/NO, ROM H ADDRESS 04 PROBABLY TRANSLATED
/AN IOT WITH BIT 3 SET AS AN OPERATE,
/OR DEVICE ON SYSTEM USES 6770 IOT,
/*****
/TEST THAT OVERFLOW F/F DOES NOT SET SKIP WHEN NOT DOING AN ISE.
2502 7000 XOR39, NOP/JMS POINT
2503 7340 IZITTS, CLA CLL CMA /AC TO 7777
2504 3010 DCA 10 /SET AUTO-INDEX LOCATION TO 7777
2505 1376 TAD (JMP I 7 /SET UP INTERRUPT RETURN
2506 3036 DCA CHNCON
2507 1373 TAD (PCTST2
2510 3007 DCA 7 /SET RETURN FROM INTERRUPT
2511 1051 TAD K200 /INTERRUPT BIT
2512 6146 LOADA /INTERRUPT
2513 6140 CNTENA
2514 5410 JMP I 10 /0010=7777 SO WE AUTO-INDEX AND JUMP TO 0000, BUT
/INTERRUPT COMES UP AT TP1 OF FETCH
/AND IS HONORED AFTER DEFER MAJOR STATE
/DID NOT INTERRUPT
2515 7402 HLT /GET INTERRUPTED PC(SHOULD BE 0000)
2516 1000 TAD 0 /DID WE END UP AT LOCATION 0000 AFTER JUMP?
2517 7440 SEA /NO, IF AC=0001 THE OVERFLOW F/F PROBABLY
2520 7402 HLT /SET THE SKIP F/F DURING DEFER OF JMP I (AUTO-INDEX)
/*****
/TEST INTERRUPT TO NOT INTERFERE WITH A NON-SKIPPING ISE
2521 1372 TAD (JMP I 0

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2522 3036 DCA CHNCON
2523 3003 DCA BIT3
2524 6001 ION
2525 1051 TAO K200
2526 6146 LOADB
2527 6140 CNTENA
2528 2003 ISZ BIT3 /INTERRUPT NEXT CYCLE
2529 7410 RETURN, SKP /SHOULD NOT SKIP
2530 7402 HLTCD, HLT /INTERRUPTED, RETURNED, AND DID NOT SKIP,
2531 1332 TAO HLTCD /INTERRUPT CAUSED ISZ TO SKIP
2532 3036 DCA CHNCON /MAKE HLT FOR UNEXPECTED INTERRUPT
2533 6001 ION

/*****
/TEST INDICATE LOGIC, FIRST MAKE IND1+2 LOW AND TEST FOR AC TO BUS
XOR40, NOP/JMS POINT
TSINDL, CLRAL /CLEAR SIMULATOR
TAO K2525 /AC TO 2525
MQL K446 /MO TO 2525, AC TO 0000
TAO K446 /IND1+2 LOW AC TO BUS
LOADB
CNTENA /ENABLE IND 1&2 TO BUS
TAO K5253 /AC TO 5253, AC ON BUS DURING TS1
READA /READ REGISTER A, AC REMAINS 5253
TAO K2525 /AC TO 0000
SZA
HLT /IND1+2 LOW DID NOT GATE AC TO BUS DURING TS1

/TEST THAT MO GATES TO BUS DURING TS1 WHEN IND1=H1, IND2=L0
XOR41, NOP/JMS POINT
TSINOB, CLRAL
TAO K2525 /MO TO 2525
MQL K444 /IND1=H IND2=L
TAO K5252 /ENABLE IND 1&2 (MO WILL BE ON BUS DURING TS1)
CNTENA /AC TO 2525
READA /AC TO 0000
TAO K5253 /WAS MO ON BUS AT TS1?
SZA

/A HALT AT THE NEXT LOCATION CAN BE CAUSED BY HAVING THE FRONT PANEL "INDICATE" SWITCH
/IN SOME POSITION OTHER THAN THE "MD", "STATUS", OR "STATE" POSITIONS. SEE
/SECTION 3, PART D.
2565 7402 HLT /NO, IND1=H1, IND2=LOW DID NOT PUT MO TO BUS
2566 5771 JMP TSINDC
2567 0444 K444, 0444
2570 0446 K446, 0446
2571 2600
2572 5400
2573 2516
2574 5353
2575 2426
2576 5407
2577 2413
2600 PAGE

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/*****
/TEST THAT IND1=H1, IND2=H1 PUTS STATUS ON BUS DURING TS1
/SINCE STATUS TO BUS IS A FUNCTION OF ROM J, THIS TEST MUST BE DONE
/8 TIMES WITH A DIFFERENT OCTAL DIGIT IN MO BITS 9 THRU 11 EACH TIME
/TO COVER ALL THE ADDRESSES USED IN ROM J TO PERFORM THIS FUNCTION
2600 3112 TSINDC, DCA TESLOC
2601 1071 TAO K7770 /-8
2602 3003 DCA BIT3
2603 7000 XOR42, NOP/JMS POINT
2604 6145 CLRAL /CLEAR SIMULATOR
2605 7120 CLL CML /SET LINK FOR STATUS INDICATION
2606 1304 TAO K440
2607 6147 LOADB
2610 6140 CNTENA /ENABLE IND OUTPUTS TO BUS
2611 1112 TAO TESLOC /TO GET DIFFERENT BITS IN MO9 THRU 11
2612 6141 READA /AC SHOULD BE 42XX
2613 0067 AND K7700 /MASK OUT IF + DF
2614 1305 TAO H4200
2615 7440 SZA

/A HALT AT THE FOLLOWING LOCATION CAN BE CAUSED BY HAVING THE FRONT PANEL
/"INDICATE" SWITCH IN SOME POSITION OTHER THAN THE "MD", "STATUS", OR
/"STATE" POSITION, SEE SECTION 3, PART D.
2616 7402 HLT /DIDN'T GET STATUS TO BUS WHEN IND1+2 H1
/IF (TESLOC) = 0000 THEN SUSPECTED FAILURE
/IN INDICATE LOGIC, IF (TESLOC) = NON-ZERO
/THEN SUSPECTED ROM J FAILURE

2617 7000 XOR43, NOP/JMS POINT
2620 2112 ISZ TESLOC /FOR NEXT COMBO OF MO9 THRU 11
2621 2003 ISZ BIT3 /ALL COMBINATIONS TRIED? (8)
2622 5203 JMP XOR42 /NO,

/TEST THAT AC, MO, OR STATUS IS NOT ON DATA BUS DURING TS1 WHEN IND1=L0, IND2=H1
XOR44, NOP/JMS POINT
TSINDD, TAO K442
LOADB /IND1=LOW IND2=H1
TAO K2525 /MO TO 2525
MQL /AC TO 7777
CLA CMA /AC TO 0000
CNTENA
READA
SZA
HLT /AC, MO, OR STATUS ON BUS WHEN IND1=LOW IND2=H1
2635 7000 XOR45, NOP/JMS POINT

/CONTINUED TEST OF ROM E
/TEST FOR INPUT, DATA ADDED TO PC WHEN C0=H, C1=H, C2=L (ROM E ADDRESS 23)
TSTCLE, XRC1 /DISABLE XOR INTERRUPTS
CLRAL /CLEAR SIMULATOR
TAO (BACKAD
DCA 7 /SET UP INTERRUPT RETURN
TAO (JMP I 7
DCA CHNCON

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2644 1131 TAO K1000
2645 6147 LOADB /C0&C1=H C2=L
2646 1001 TAO K200
2647 6146 LOADA /INTERRUPT, 1 CYCLE DELAY
2650 6140 CNTENA
2651 6142 STROB /INTERRUPT AFTER THIS INSTRUCTION
2652 7402 HLT /DID NOT INTERRUPT OR ADD TO PC
2653 1074 BACKAD, TAO SAVAC /GET INTERRUPTED AC
2654 7440 SEA /STILL ZERO?
2655 7402 HLT /NO, AC WAS CHANGED,
2656 1255 TAO ,=1
2657 3036 DCA CHNCON /SET UP FOR UNEXPECTED INTERRUPT
2660 1000 TAO 0 /GET INTERRUPTED PC
2661 6001 IOV
2662 1375 TAO (-BACKAD-200+1
2663 7440 SEA /WAS PC CORRECT?
2664 7402 HLT /NO, C0&1=L AND C2=H DID NOT ADD TO PC
/*****
/A FAILURE IN THE TEST ABOVE OR THE TEST BELOW COULD BE DUE TO A PROBLEM
/IN THE "NOT LAST XFER" OR "I/O STALL" LOGIC. THESE TWO ROUTINES ARE THE
/ONLY ONES IN THE PROGRAM THAT CAUSE THE SIMULATOR TO ASSERT "NOT LAST XFER",
/WHICH IN TURN ASSERTS I/O STALL IN THE CPU.
/*****
/TEST FOR INPUT, DATA ADDED TO PC WHEN C0&2=L AND C1=H (ROM E ADDRESS 22)
XOR49, NOP/JMS POINT
TSTCLF, CLRAL /CLEAR SIMULATOR
TAO (BAKADB
DCA 7
TAO K5000
LOADB /C0&C2=L C1=H
TAO (JMP I 7
DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
TAO K200 /INTERRUPT, 1 CYCLE DELAY
LOADA
CNTENA
STROB /GATE C LINES, INTERRUPT AFTER THIS INSTRUCTION
HLT /DID NOT ADD TO PC OR INTERRUPT
K442, 0442
K5000, 5000
K440, 0440
M4200, 3600
BAKADB, TAO SAVAC
SEA
HLT /AC WAS AFFECTED BY C0&2=L C1=H
TAO ,=1
DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
TAO 0 /INTERRUPT, 1 CYCLE DELAY
IOV
TAO (-K442-200+1
SEA /WAS PC CORRECT?
HLT /NO, C0&2=L C1=H DID NOT ADD TO PC
/*****
/TEST FOR INPUT TO PC WHEN C0=H C1&2=L (ROM E ADDRESS 21)
XOR50, NOP/JMS POINT
TSTCLG, XRSI /ENABLE XOR INTERRUPTS

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2722 6145 CLRAL
2723 1372 TAO (BAKADC
2724 3007 DCA 7 /SET UP INTERRUPT RETURN
2725 1336 TAO K3000
2726 6147 LOADB /C0=H C1&C2=L
2727 1376 TAO (JMP I 7
2730 3036 DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
2731 1051 TAO K200
2732 6146 LOADA /INTERRUPT, 1 CYCLE DELAY
2733 6140 CNTENA
2734 6142 STROB /GATE C LINES, INTERRUPT AFTER INSTRUCTION
2735 7402 HLT /DID NOT INTERRUPT OR LOAD PC
2736 3000 K3000, 3000
2737 1074 BAKADC, TAO SAVAC
2740 7440 SEA /GET INTERRUPTED AC
2741 7402 HLT /STILL 0000?
2742 1341 TAO ,=1 /NO, AC WAS CHANGED WHEN C0=H C1&2=L
2743 3036 DCA CHNCON /SET UP FOR UNEXPECTED INTERRUPT
2744 1000 TAO 0 /GET INTERRUPTED PC
2745 6001 IOV
2746 1366 TAO M200 / 7600
2747 7440 SEA /PC CORRECT?
2750 7402 HLT /NO, C0=H C1&2=L DID NOT LOAD PC
/*****
/TEST FOR INPUT TO PC WHEN C0,C1, & C2 = L (ROM E ADDRESS 20)
XOR51, NOP/JMS POINT
TSTCLH, CLRAL /CLEAR SIMULATOR
TAO (BAKADD
DCA 7 /SET UP INTERRUPT RETURN
TAO K7000
LOADB /C0,C1,&C2=L
TAO (JMP I 7
DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
TAO K200 /INTERRUPT, 3 CYCLE DELAY
LOADA
CNTENA
STROB /GATE C LINES, INTERRUPT AFTER THIS INSTRUCTION
HLT /DID NOT INTERRUPT OR INPUT TO PC
M200, 7600
2766 7600
2771 3000
2772 2737
2773 4677
2774 2706
2775 4726
2776 5407
2777 2653
3000 PAGE
3001 1074 BAKADD, TAO SAVAC
3002 7440 SEA /GET INTERRUPTED AC
3003 7402 HLT /STILL 0000?
3004 1202 TAO ,=1 /NO, AC WAS LOADED WHEN ALL C LINES WERE LOW
3005 3036 DCA CHNCON /SET UP FOR UNEXPECTED INTERRUPT
3006 1000 TAO 0

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3006 6001 ION
3007 1777 TAO K200 / 7600
3010 7440 SZA /WAS PC CORRECT?
3011 7402 HLT /NO, 00,1,8, LOW DID NOT INPUT TO PC
/*****
/TEST ROM 3, ADDRESS 15 BY INTERRUPTING AT THE END OF A DEFER
XOR92, NOP/JMS POINT
ROMB15, CLRAL /CLEAR SIMULATOR
3012 7000 TAO (ROBRET
3013 6145 DCA 7 /SET UP INTERRUPT RETURN
3014 1376 TAO (JMP I 7
3015 3007 DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
3016 1375 TAO K201
3017 3036 LOADA /INTERUPT, 3 CYCLE DELAY
3020 1227 CNTENA
3021 6146 CLA GLL
3022 6140 JMP I (-, +1
3023 7300 /INTERUPT AFTER DEFER
3024 5774 ROBHLT, HLT /DID NOT INTERRUPT
3025 7402 HLT /DID NOT INTERRUPT, BUT CARRY IN ENABLED
3026 7402 K201, 0201
3027 0201 ROBRET, TAO =2
3030 1226 DCA CHNCON /SET UP FOR UNEXPECTED INTERRUPT
3031 3036 TAO 0 /GET INTERRUPTED PC
3032 1000 ION
3033 6001 TAO (=ROBHLT
3034 1373 SZA
3035 7440 HLT /WAS PC CORRECT?
3036 7402 HLT /NO, INTERRUPT AFTER DEFER DID NOT STORE CORRECT PC
/*****
/TEST "ROM ADD L" TO DISABLE STORING PC FOR A JMS AND DISABLE MA+1 TO PC,
XOR93, NOP/JMS POINT
TSROMA, CLRAL /CLEAR SIMULATOR
3037 7000 TAO (GORETU
3038 6145 DCA 7 /SET UP INTERRUPT RETURN
3039 1372 TAO (JMP I 7
3040 3007 DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
3041 1375 TAO ROBHLT
3042 3036 DCA ROMCLR /SET UP INDICATOR
3043 1225 TAO K221
3044 3254 LOADA /CAUSE PULSE ON ROM ADD L, 3 CYCLE DELAY, & INT
3045 1225 CNTENA
3046 1357 NOP
3047 1357 JMS =+1
3048 6146 /PULSE ON ROM ADD L SHOULD ARRIVE DURING
3049 6140 /EXECUTE AND DISABLE STORING PC AND DISABLE
3050 7000 /MA+1 TO PC, IN CORE MACHINES NEXT LOCATION IS CLEARED SINCE
3051 4254 /ROM ADD L ARRIVES AND PREVENTS RESTORING MEMORY
3052 7000
3053 4254
3054 7402 ROMCLR, HLT
3055 7402 HLT /DID NOT INTERRUPT AFTER EXECUTE OF JMS
3056 1225 GORETU, TAO ROBHLT
3057 3036 DCA CHNCON /SET UP FOR UNEXPECTED INTERRUPT
3058 1000 TAO 0 /GET INTERRUPTED PC
3059 6001 ION
3060 1000 TAO (=ROMCLR
3061 6001 SZA
3062 1371 HLT /DID ROM ADD L DISABLE MA+1 TO PC?
3063 7440 /NO, ROM ADD L DID NOT DISABLE MA+1 TO PC
3064 7402
/*****

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/TEST BREAK FROM MEMORY AND BREAK BACK INTO MEMORY - (USES ROMA ADDRESSES 30 THRU 37
/AND ROM B ADDRESS 3)
XOR94, NOP/JMS POINT
DBOUT, CLRAL
3065 7000 TAO K100
3066 6145 LOADB /BREAK OUT
3067 1050 CLA CMA /AC TO 1777
3068 6147 CDF 00 /IN CASE PROGRAM IS RUNNING IN EXT MEMORY
3069 7240 DCA I K1777 /1777 TO 7777, AC TO 0000
3070 6201 SETBK /SET BREAK REQUEST
3071 3454 CNTENA
3072 6143 NOP
3073 6140 READA
3074 6143 CMA
3075 6140 SZA
3076 7000 HLT
3077 6141 /AC TO 1777
3078 7240 /AC TO 0000 ANY BITS DROPPED FROM DB ARE SET
3079 7440 /ANY BITS DROP DURING DB?
3080 7402 /YES, DB OUT DID NOT WORK, ANY BIT SET IN
3081 7402 /AC IS A BIT DROPPED ON A DB OUT OF MEMORY.
/*****
/TEST BREAK INTO MEMORY (ADDRESS 1777)
TSBKTO, DCA I K1777 /CLEAR 1777
3103 3454 LOADB /BREAK IN
3104 6147 SETBK /BREAK IN OCCURS END OF NEXT CYCLE
3105 6143 CNTENA
3106 6140 NOP
3107 7000 TAO I K1777
3108 1454 /AC TO 1777 IF BREAK IN AND OUT BOTH WORKED
3109 7040 /AC TO 0000, DROPPED BITS APPEAR AS 1'S
3110 7440 /ANY BITS DROPPED DURING DATA BREAK?
3111 7402 /YES, BREAK OUT OR BREAK IN FAILED COMPLETELY
3112 7402 /OR DROPPED BITS, BITS NOT TRANSFERRED ARE
3113 7402 /NOW SET IN AC
/*****
/TEST DATA BREAK ADD TO MEMORY
TSBKAD, TAO K200
3114 1051 LOADB
3115 6147 TAO K5252
3116 1064 DCA I K1777 /SET 1777 TO 5252
3117 3454 SETBK /ST BREAK REQUEST
3118 6143 CNTENA /BREAK IN AND ADD TO 1777 NEXT CYCLE
3119 7000 NOP /1777 TO 5251 DURING DATA BREAK
3120 6143 TAO I K1777 /AC TO 5251
3121 6140 TAO K2527 /AC TO 0000
3122 7000 SZA /DID ADD TO MEMORY BREAK WORK?
3123 1454 HLT /NO, DATA BREAK ADD TO MEMORY DID NOT FUNCTION.
3124 1360 /ROM A ADDRESSES 30 THRU 33 ARE SUSPECT
3125 7440
3126 7402
/*****
/TEST BREAK OUT OF ADDRESS 0017
XOR95, NOP/JMS POINT
BKFLOW, CLRAL
3127 7000 TAO K2525 /AC TO 2525
3128 6145 DCA I K17 /00017 TO 2525 (DF#0)
3129 1056 CLA GLL IAC RTL /+4
3130 3761 LOADA /BREAK FORM LOW ADDRESS (0017)
3131 7307 TAO K100
3132 6146 /BREAK OUT OF MEMORY
3133 1050 LOADB
3134 6147 SETBK
3135 1050
3136 6147
3137 6143

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3140 6140 CNTENA
3141 7000 NOP
3142 6141 READA
3143 1065 TAD K9253
3144 7440 SZA
3145 7402 HLT
/BREAK AFTER NOP, 2525 TO SIMULATOR
/AC TO 2525
/BREAK FROM LOW ADDRESS WORK?
/NO, BREAK OUT OF ADDRESS 0017 FAILED, CPMA
/BIT MAY NOT BE GOING TO ITS HIGH IMPEDENCE STATE.

/*****
/TEST BREAK INTO 0017
BKIN17, LOADB /BREAK IN
DCA I K17 /CLEAR 00017
SETBK
CNTENA
CLA CMA /BREAK AFTER THIS INSTRUCTION
CMA /AC TO 0000
SEA /DATA BREAK AFFECT AC?
HLT /YES, DATA BREAK AFFECTED AC.
JMP BKINTO
K221, 0221
K2527, 2527
K17, 0017

3146 6147
3147 3761
3150 6143
3151 6140
3152 7240
3153 7040
3154 7440
3155 7402
3156 5770
3157 0221
3160 2527
3161 0017
3170 3200
3171 4724
3172 3056
3173 4753
3174 3025
3175 5407
3176 3030
3177 2766
3200 1773
3201 1065
3202 7440
3203 7402
PAGE
BKINTO, TAD I K178 /AC TO 2525
TAD K9253 /AC TO 0000
SZA /BREAK INTO 0017 CORRECTLY
HLT /NO, BREAK INTO 0017 FAILED.

/*****
/CAUSE AN ASYNCHRONOUS DATA BREAK
XOR56, NOP/JMS POINT
ASYNCB, CLRAL
TAD K400 /ASYNC DATA BREAK
LOADA /AC TO 7777
CLA CMA /1777 TO 7777
DCA I K1777
SETBK
CNTENA /BREAK WITHIN 14 USEC.
CLA CMA CLL RTL /-3
DCA TESLOC
ISE TESLOC
JMP -1 /TIME WASTER, DB OCCURS DURING THIS WAIT.
TAD I K1777 /AC TO 0000
SZA /BREAK OCCUR CORRECTLY?
HLT /NO, ASYNC DATA BREAK DID NOT CLEAR 1777

/TEST FOR PROPER OPERATION OF 5 BREAKS IN A ROW
XOR57, NOP/JMS POINT
TS5BRK, CLRAL
3223 7000
3224 6145

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3225 1050 TAD K100
3226 6147 LOADB
3227 1056 TAD K2525
3230 3454 DCA I K1777 /((1777) TO 2525
3231 6143 SETBK
3232 6140 CNTENA /((1777) TO SIMULATOR
3233 7000 NOP
3234 1043 TAD K4 /LOW ADDRESS FOR DB
3235 6146 LOADA
3236 1041 TAD K1 /5 BREAKS
3237 6147 LOADB
3240 1371 TAD M98
3241 3112 DCA TESLOC /SET UP LOOP OF 5
3242 1271 TAD K12
3243 3010 DCA 10 /POINTER FOR ADDRESS 13
3244 3410 DCA I 10 /CLEAR 13 THRU 17
3245 2112 ISE TESLOC /ALL 5 CLEARED?
3246 5244 JMP -2 /NO
3247 6143 SETBK
3250 6140 CNTENA
3251 7000 NOP /((13) THRU (17) TO 2525
3252 1271 TAD K12
3253 3010 DCA 10
3254 1371 TAD M98
3255 3112 DCA TESLOC
3256 1410 SOMMOR, TAD I 10 /GET A LOCATION BETWEEN 13 AND 17
3257 1065 TAD K9253 /COMPARE TO 2525
3260 7440 SZA /EQUAL 2525?
3261 7402 HLT /NO, 5 BREAKS DID NOT WORK CORRECTLY
3262 2112 ISE TESLOC /ALL LOCATIONS CHECKED?
3263 5256 JMP SOMMOR /NO

/*****
/TEST THAT INITIALIZE LINE(CR1) IS NOT OPEN TO BUS
XOR58, NOP/JMS POINT
TSINLN, CLRAL /CLEAR SIMULATOR
CLA CMA /AC TO 7777
DCA TESLOC
ISE TESLOC /SHOULD SKIP AND SET OVERFLOW F/F IN SIMULATOR
K12, 0012 /SKIP ON OVERFLOW IN SIMULATOR, SHOULD SKIP
SKPOFV /OVERFLOW DID NOT SET OVERFLOW F/F IN SIMULATOR,
HLT /CLEAR OVERFLOW F/F IN SIMULATOR
CAF /SKIP ON OVERFLOW IN SIMULATOR, SHOULD NOT SKIP NOW.
SKPOFV /DID NOT SKIP, OK,
SKP /CAF DID NOT CLEAR OVERFLOW F/F IN SIMULATOR,
HLT

/TEST THAT DMA DISABLES CLOCKING INTERRUPT CIRCUITRY
XOR59, NOP/JMS POINT
TSDMIT, CLRAL
ION
TAD K2525
DCA I K1777
TAD K100
LOADB /BREAK OUT OF MEMORY (1777)
IAG

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3310 6146 LOADA /DELAY OF THREE CYCLES
3311 6143 SETBK /SET BREAK REQUEST
3312 6140 CNTENA /BREAK IN 3 CYCLES
3313 7000 NOP
3314 3454 DCA I K1777 /DB SHOULD OCCUR AT BEGINNING OF EXECUTE, BEFORE
/1777 IS CLEARED, THEREFORE DB DATA SHOULD BE 2525,
3315 6141 READA /READ DB DATA, AC TO 2525
3316 1065 TAD K5253 /AC TO 0000
3317 7440 SEA /DB DATA CORRECT?
3320 7402 HLT /NO, IF AC = 5253, THEN DB OCCURRED AFTER THE EXECUTE
/OF THE DCA, OR DID NOT OCCUR, IF AC NOT EQUAL
/2523 THEN DATA BREAK INTERFERED WITH THE DCA

3321 1051 TAD K200
3322 7001 IAS
3323 6146 LOADA
3324 6147 LOADB /INTERUPT DELAY OF THREE CYCLES
3325 1377 TAD (DMARET
3326 3007 DCA 7 /SET UP INTERRUPT RETURN
3327 1376 TAD (JMP I 7
3330 3036 DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
3331 6143 SETBK /SET BREAK REQUEST
3332 6140 CNTENA /ENABLE INT AND DB ON THIRD TP1
3333 7000 NOP /CYCLE MASTER
3334 1454 TAD I K1777 /DB OCCURS JUST BEFORE EXECUTE OF TAD AND LOADS
/ADDRESS 1777 WITH 2525, THEN THE EXECUTE OF THE TAD
/LOADS THE AC WITH 2525, INTERRUPT OCCURS AFTER
/THE EXECUTE OF THE TAD,
/3335 7402 HLT /DID NOT INTERRUPT CORRECTLY, DATA BREAK AND INTERRUPT INTERFERED.
3336 1074 DMARET, TAD SAVAC /GET INTERRUPTED AC, AC TO 2525.
3337 7450 SNA /HAS AC CLEAR BECAUSE THE INTERRUPT CIRCUITRY WAS
/LOCKED DURING DB, PREVENTING THE EXECUTE OF THE TAD?
/YES, INTERRUPT WAS CLOCKED DURING DB.
3340 7402 HLT /AC TO 0000.
3341 1065 TAD K5253 /WAS AC CORRECT?
3342 7440 SEA /NO, DMA AND INTERRUPT INTERFERED,
3343 7402 HLT /SET UP FOR UNEXPECTED INTERRUPTS
3344 1343 TAD ,=1
3345 3036 DCA CHNCON
3346 6001 IOV

/*****
/TEST ADDRESSES 10 THRU 13 OF ROM A.(DEPOSIT NON-STOP FUNCTION)
3347 7000 XOR60, NOP/JMS POINT /NOP FOR PDP-8/E TYPE COMPUTERS
3348 7410 TSDPNS, SKP /USED TO SKIP DEPOSIT NON-STOP TEST FOR PDP-8/E
3349 5775 JMP XOR62 /CLEAR OUT SIMULATOR
3350 6145 CLRAL
3351 7240 CLA CMA
3352 3454 DCA I K1777
3353 1374 TAD (ROMRTN
3354 3007 DCA 7 /SET UP INTERRUPT RETURN
3355 1376 TAD (JMP I 7
3356 3036 DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
3357 1372 TAD K220
3360 6147 LOADB /DMA WITH LA ENABLE LOW, KEY CONTROL HIGH,
/AND BREAK DATA CONTROL LOW.
3363 1051 TAD K200
3364 6146 LOADA /INTERUPT ENABLE

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3365 6143 SETBK
3366 6140 CNTENA
3367 1454 TAD I K1777 /DMA WILL DEPOSIT 0 IN LOCATION 1777 BEFORE THE DEFER
/OF THE TAD, THEREFORE THE AC WILL BE LOADED WITH
/0000 DURING THE EXECUTE OF THE TAD, THE DMA WILL LOAD
/THE PC WITH 2000, THEREFORE THE MA=2000
/AFTER THE TAD, AN INTERRUPT OCCURS IMMEDIATELY
/AFTER THE COMPLETION OF THE TAD,
/FAILED TO LOAD PC OR INTERRUPT,

3370 7402 HLT
3371 7773 M5B, 7773
3372 0220 K220, 0220
3373 0017 K17B, 0017
3374 3400
3375 3415
3376 3407
3377 3336
3400 3400 PAGE
3401 1074 ROMRTN, TAD SAVAC /GET INTERRUPTED AC, AC TO 0000 IF DMA WORKED
3402 7440 SEA /DMA WORK CORRECTLY?
3403 7402 HLT /NO, DEPOSIT NON-STOP FAILED, ROM A ADDRESSES 10-13.
3404 1202 TAD ,=1
3405 3036 DCA CHNCON
3406 1106 TAD SAVFLD /SET UP BACK TO WHATEVER IT WAS,
3407 6005 RTF /RESTORE
3410 7300 CLA CLL
3411 1000 TAD 0 /GET INTERRUPTED PC, AC TO 2000
3412 5212 JMP ,=1 /ENABLES INTERRUPTS
3413 1377 TAD (=2000 /AC TO 0000
3414 7402 SEA /WAS PC LOADED BY DMA?
HLT /NO, DEPOSIT NON-STOP DID NOT LOAD PC
/EXPECTEDLY, ROM A ADDRESSES 10 THRU 13 AT FAULT.

/*****
/TEST THAT MA,MS LOAD CONTROL LOW INHIBITS PC TO CPMA
3415 7000 XOR62, NOP/JMS POINT
3416 6145 TSMMS, CLRAL
3417 1055 TAD K2000 /PULSE MA,MS LOAD CONTROL LINE
3420 6146 LOADA /PULSE WILL ARRIVE DURING NEXT INSTRUCTION
3421 6140 CNTENA /THIS INSTRUCTION WILL BE EXECUTED TWICE, CAUSING A FINAL
3422 7001 IAS /AC OF 0002, THE MA,MS LOAD CONTROL PULSE DURING
/THE FIRST EXECUTION, PREVENTS PC TO CPMA AND
/THREBY CAUSES THE SAME INSTRUCTION TO BE EXECUTED OVER,
/AC TO 0000
/NO, MA,MS LOAD CONTROL DID NOT INHIBIT PC TO CPMA

/*****
/TEST REMAINDER OF OMNIBUS LINES THAT SO FAR HAVE NOT BEEN USED IN BOTH A
/HIGH AND LOW STATE: MA0,RUN,INT IN PH0G,LINK L, LINK DATA L,F SET L,
/PETCH,EXECUTE,DEFER,IR0,IR1,IR2,
3426 7000 XOR63, NOP/JMS POINT
3427 6145 TSLINE, CLRAL /AC TO 0002
3430 7005 IAC RAL /5 CYCLE DELAY
3431 6146 LOADA
3432 7001 IAC
3433 6140 CNTENA /ENABLE STATUS READING

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3434 7200 CLA /1ST CYCLE OF DELAY
3435 7000 NOP /2ND CYCLE
3436 0441 AND I K4000 /3TH CYCLE IS EXECUTE OF THIS INSTRUCTION
3437 6141 READA /AC TO 6120
3440 7500 SMA /HAS MA0 LOW?
3441 7402 HLT /NO, MA0 STUCK HIGH ON BUS
3442 7006 RTL /RUN BIT TO LINK.
3443 7420 SNL /RUN LOW?
3444 7402 HLT /NO, RUN LINE STUCK HIGH ON BUS.
3445 7500 SMA /INT IN PROG LOW?
3446 7402 HLT /NO, INT IN PROG STUCK HIGH ON BUS
3447 7006 RTL /
3450 7430 SEL /LINK LINE HIGH?
3451 7402 HLT /NO, LINK LINE(AV2) STUCK LOW ON BUS,
3452 7510 SPA /LINK DATA LINE HIGH?
3453 7402 HLT /NO, LINK DATA LINE(CR2) STUCK LOW ON BUS,
3454 7006 RTL /
3455 7420 SNL /F SET LOW?
3456 7402 HLT /NO, F SET(DP2) STUCK HIGH ON BUS,
3457 7510 SPA /FETCH LINE HIGH?
3460 7402 HLT /NO, FETCH LINE(DJ2) STUCK LOW ON BUS,
3461 7006 RTL /
3462 7420 SNL /EXECUTE LOW?
3463 7402 HLT /NO, EXECUTE(DL2) STUCK HIGH ON BUS,
3464 7510 SPA /DEFER HIGH?
3465 7402 HLT /NO, DEFER(DK2) STUCK LOW ON BUS,
3466 7006 RTL /
3467 7430 SEL /IR0 HIGH?
3470 7402 HLT /NO, IR0 STUCK LOW ON BUS,
3471 7510 SPA /IR1 HIGH?
3472 7402 HLT /NO, IR1 STUCK LOW ON BUS,
3473 7004 RAL /
3474 7710 SPA CLA /IR2 HIGH?
3475 7402 HLT /NO, IR2 STUCK LOW ON BUS.
/*****
/TEST FOR IR0,IR1,AND IR2 LOW, FETCH LOW, MA0 HIGH, EXECUTE HIGH AND LINK LOW
XOR64, NOP/JMS POINT
TSLINB, CLRAL
3476 7000 IAC GML /AC TO 0001, LINK TO 1
3477 6145 CNTENA /ENABLE STATUS READING
3500 7021 CLA /STATUS LOADED AT TP3 OF THIS INSTRUCTION
3501 6140 READA /READ SIMULATOR STATUS REGISTER
3502 7200 SPA /MA0 HIGH?
3503 6141 HLT /NO, MA0 STUCK LOW ON BUS,
3504 7510 RTL /
3505 7402 RTL /
3506 7006 SNL /LINK LOW?
3507 7006 HLT /NO, LINK LINE(AV2) STUCK HIGH ON BUS,
3510 7420 RTL /
3511 7402 SMA /FETCH LOW?
3512 7006 HLT /NO, FETCH (DJ2) STUCK HIGH ON BUS,
3513 7500 RTL /
3514 7402 SEL /EXECUTE HIGH?
3515 7006 HLT /NO, EXECUTE(DL2) STUCK LOW ON BUS,
3516 7430 RTL /
3517 7402 RTL /
3520 7006 RTL /

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3521 7420 SNL /IR0 LOW?
3522 7402 HLT /NO, IR0(DD2) STUCK HIGH ON BUS.
3523 5775 JMP TSLNCN
3575 3600
3576 3741
3577 6000
3600 3600 PAGE
3601 7500 TSLNCN, SMA /IR1 LOW?
3601 7402 HLT /NO, IR1(DE2) STUCK HIGH ON BUS,
3602 7004 RAL
3603 7700 SMA CLA /IR2 LOW?
3604 7402 HLT /NO, IR2(DH2) STUCK HIGH ON BUS.
/*****
/TEST FOR LINK DATA LOW ON BUS,
XOR65, NOP/JMS POINT
TSLINC, CLRAL
3605 7000 CLA GLL IAC RAL /+2
3606 6145 LOADA /5 CYCLE DELAY
3607 7305 IAC
3610 6146 CNTENA /ENABLE STATUS READING
3611 7001 RTR /AC TO 4000
3612 6140 TAD SAVFLD /GET IF AND DF
3613 7012 NOP
3614 1106 RTF /STATUS READ BY SIMULATOR AT TP3 THIS INSTRUCTION
3615 7000 CLA
3616 6005 READA /READ STATUS
3617 7200 AND K200
3620 6141 SNA CLA /LINK DATA LINE LOW?
3621 0051 HLT /NO, LINK DATA LINE(CR2) STUCK HIGH ON BUS
3622 7650
3623 7402
/*****
/TEST FOR DEFER LOW ON BUS AND F SET HIGH
XOR66, NOP/JMS POINT
TSLIND, CLRAL
3624 7000 IAC /DELAY OF 3 CYCLES
3625 6145 LOADA
3626 7001 IAC
3627 6146 CNTENA
3630 7001 CLA
3631 6140 AND I K200 /READ STATUS AT TP3 OF DEFER
3632 7200 READA
3633 0451 AND K10
3634 6141 SNA CLA /HAS DEFER LOW?
3635 0044 HLT /NO, DEFER LINE(DK2) STUCK LOW ON BUS,
3636 7650 READA /READ STATUS AGAIN,
3637 7402 AND K100 /MASK OUT F SET,
3640 6141 SEA /F SET HIGH?
3641 0050 HLT /NO, F SET STUCK LOW ON BUS (DP2)
3642 7440
3643 7402
/*****
/TEST FOR INT IN PROG HIGH
XOR67, NOP/JMS POINT
TSLINF, CLRAL
3644 7000 TAD K200
3645 6145 IAC /DELAY OF THREE CYCLES
3646 1051 LOADA
3647 7001 TAD KJMPI0
3650 6146
3651 1334

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3652 3036 DCA CHNCON
3653 7001 IAC
3654 6140 CNTENA /ENABLE STATUS CHECKING
3655 7200 CLA
3656 7000 NOP /INTERRUPT AT END OF THIS INSTRUCTION
3657 1265 TAO KHALT /RETURN HERE FROM INTERRUPT
3660 3036 DCA CHNCON /SET UP FOR UNEXPECTED INTERRUPT
3661 6001 IOV
3662 6141 READA /READ STATUS FROM SIMULATOR
3663 0131 AND /MASK OUT INT IN PROG
3664 7650 SNA CLA /INT IN PROG HIGH?
3665 7402 KHALT, HLT /NO, INT IN PROG STUCK LOW ON BUS, (BP2)
/*****
/TEST THAT CPMA DISABLE LOW DISABLES MA+1 TO PC
XOR68, NOP/JMS POINT
TSCPD5, CLRAL
3666 7000 TAO KCPMRE
3667 6145 DCA 7 /SET UP INTERRUPT RETURN
3670 1336 TAO K4201
3671 3007 TAO /CPMA DISABLE PULSE, INTERRUPT, 3 CYCLE DELAY
3672 1342 LOADA
3673 6146 TAO KJMP17
3674 1335 DCA CHNCON /SET UP FOR EXPECTED INTERRUPT
3675 3036 CNTENA
3676 6140 NOP
3677 7000 JMS ,+1 /CYCLE MASTER
3700 4301 /CPMA DISABLE PULSE OCCURS DURING EXECUTE
/*****
3701 0000 CPSTOR, 0 /PREVENTING MA+1 TO PC
3702 7402 HLT
3703 1302 CPMRET, TAO ,+1 /DID NOT INTERRUPT
3704 3036 DCA CHNCON
3705 1000 TAO 0 /SET UP FOR UNEXPECTED INTERRUPT
3706 6001 IOV
3707 1333 TAO KCPSTO
3710 7440 SZA
3711 7402 HLT /HAS PC AS EXPECTED?
/NO, CPMA DISABLE DID NOT PREVENT MA+1 TO PC
/*****
/TEST FOR HALT AT END OF PASS
XOR69, NOP/JMS POINT
FINALE, DCA PARTHO /GO BACK TO REGULAR TYPE POWER FAIL RECOVERY
3712 7000 TAO 21 /GET HARDWARE DESIGNATOR
3713 3100 SNA CLA /USE FRONT PANEL?
3714 1021 JMP ,+3 /NO, USE PSEUDO SR,
3715 7700 OSR /YES, USE FRONT PANEL SR,
3716 5321 SKP
3717 7404 TAO 20 /GET PSEUDO SR,
3720 7410 AND K400 /MASK OUT STOP BIT,
3721 1020 SZA CLA /HALT ON PASS COMPLETE?
3722 0052 7402/HLT /YES,
3723 7640 XOR70, NOP/JMS POINT
3724 7402 TAO 21 /GET HARDWARE DESIGNATOR
3725 7000 AND K100 /MASK OUT XOR BIT
3726 1021 SZA CLA /RUNNING ON XOR?
3727 0050 JMP I XRPONT/CLA /YES, (CLA IN 1K VERSION)
3730 7640 LEAVE, JMP I KSTOVR /NO, DO TEST AGAIN.
3731 5740 KCPSTO, -CPSTOR
3732 5737
3733 4077

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3734 5400 KJMP10, JMP I 0
3735 5407 KJMP17, JMP I 7
3736 3703 KCPMRE, CPMRET
3737 0222 KSTOVR, TSCACL/0200 FOR 1K VERSION
3740 4000 XRPONT, XORSTT/(LEAVE) FOR 1K VERSION
3741 7776 M2, 7776
3742 4201 K4201, 4201
3743 7773 M5, 7773
IFDEF ACTBE <
/THIS CODE IS USED ON THE ACT-B/E LINE ONLY
ACTCHK, TAO 22 /GET HARDWARE WORD #2
RTL /QUICK VERIFY BIT TO AC0
SPA CLA /QUICK VERIFY?
JMP ACTBAK /YES, REPORT SUCCESSFUL PASS,
/NOT QUICK VERIFY, MUST TIME OUT 10 MINUTES
CLL
ISZ ACTNTR /BUMP TIMER
JMP I K200 /NO OVERFLOW, DO ANOTHER PASS.
JMP ACTBAK /OVERFLOW, REPORT GOOD RUN.
/10 MINUTES WORTH OF CPU TEST>
ACTNTR, 3240
IFDEF ACTBA <
IFDEF ONEKP2<
ACTSLO, 7777
/THIS CODE IS USED ON THE ACT-B/A LINE ONLY
ACTBKK, CLL
TAO KSTOVR
AND K77
SZA CLA /RUNNING 1K?
JMP ACTBAK /NO, REPORT IMMEDIATELY,
ISZ ACTSLO
JMP I KSTOVR /SLOW DOWN ACT REPORTING
TAO K7700
DCA ACTSLO
JMP ACTBAK>>>
/*****
/XOR INITIALIZATION CODE
4000 0000 XORSTT, 0
4001 6173 STIP /MUT POWER ON?
4002 7410 SKP /NO, DO CORE SWAP AND INITIALIZATION PASS
4003 5217 JMP XORNIT /YES, SKIP INITIALIZING AND CORE CHANGING
/*****
/START INITIALIZING THE MUT AND THE KGM
/AND CHANGE PROGRAM FOR XOR
4004 6170 XORN
4005 1377 TAO (XORTAB /POWENS MUT AND BEGINS INITIALIZATION
4006 3112 DCA TESLOC
4007 1512 DOMORE, TAO I TESLOC
4010 7450 SNA
4011 5451 JMP I K200 /ALL DONE?
4012 3116 DCA POINTC /YES, DO ONE PASS OF TEST FOR SYNC PURPOSES
4013 1376 TAO (JMS POINT /NO, SAVE POINTER,
/GET XOR HOOK INSTRUCTION

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4014 3516      DCA I POINTC
4015 2112      ISZ   TESLOC
4016 5207      JMP    DOMORE
          /ENABLE "XOR"ING
          XORNIT, CAF
4017 6007      CAF
4020 7621      CA4
4021 6171      SKXR
4022 6170      XRON          /START "XOR"ING PROCESS
4023 6007      CAF
4024 7621      CA4
4025 1375      TAD    (XORNIT+4
4026 3173      DCA    POINTX
4027 4156      JMS    POINT          /CHECK FOR STATIC ERROR
                                      /IF A STATIC ERROR EXISTS, RETURN IS MADE
                                      /TO (XORNIT+4)
                                      /TURN ON XOR INTERRUPT, NO STATIC ERROR.
                                      /BEGIN TEST.

4030 6174      XRSI
4031 5451      JMP I K200
          /*****
          /XOR HOOK TABLE
          XORTAB, EXHLT1          /THESE LOCATIONS ARE CHANGED TO
                                      /JMS POINT FOR XOR USE

4032 0221      XOR03
4033 0200      XOR04
4034 0266      XOR05
4035 0320      XOR06
4036 0346      XOR07
4037 0423      XOR08
4040 0447      XOR09
4041 0503      XOR10
4042 0530      XOR11
4043 0600      XOR12
4044 0643      XOR13
4045 1343      XOR14
4046 0667      XOR15
4047 1674      XOR16
4050 0752      XOR17
4051 1020      XOR18
4052 1056      XOR19
4053 1117      XOR20
4054 1167      XOR21
4055 1206      XOR22
4056 1241      XOR23
4057 1335      TRNXP
4060 1361      TSTFBE
4061 1714      XOR27
4062 2200      XOR28
4063 2236      XOR29
4064 1137      XOR30
4065 1745      XOR31
4066 1767      XOR32
4067 2312      XOR33
4070 2332      XOR34
4071 2400      XOR35
4072 2414      XOR36
4073 2435      XOR37
4074 2443      XOR38
4075 2456

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4076 2502      XOR39
4077 2536      XOR40
4100 2552      XOR41
4101 2603      XOR42
4102 2617      XOR43
4103 2623      XOR44
4104 2635      XOR45
4105 2262      XOR46
4106 2275      XOR47
4107 2470      XOR48
4110 2665      XOR49
4111 2720      XOR50
4112 2751      XOR51
4113 3012      XOR52
4114 3037      XOR53
4115 3065      XOR54
4116 3127      XOR55
4117 3204      XOR56
4120 3223      XOR57
4121 3264      XOR58
4122 3300      XOR59
4123 3347      XOR60
4124 3415      XOR61
4125 3426      XOR62
4126 3476      XOR63
4127 3605      XOR64
4130 3624      XOR65
4131 3644      XOR66
4132 3666      XOR67
4133 3712      XOR68
4134 3725      XOR69
4135 0000      XOR70
          0          /END OF XOR TABLE

4175 4023
4176 4156
4177 4032
          FIELD 0

```

```
0000 11111111 11011111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0100 11111111 11111111 11111111 11111111 11000000 00000011 11111111 11111111 11111111

0200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

0400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

0600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
0700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

1000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

1200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

1400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

1600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
1700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11110001

2000
2100

2200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2300 11111111 11111111 11111111 11111111 11111111 11111111 11111100 00000000 11111111

2400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2500 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

2600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
2700 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 01111111

3000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3100 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11000000 11111111

3200 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3300 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111

3400 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3500 11111111 11111111 11110000 00000000 00000000 00000000 00000000 00000000 00000111

3600 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
3700 11111111 11111111 11111111 11111111 11110000 00000000 00000000 00000000 00000000
```

```
4000 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
4100 11111111 11111111 11111111 11111100 00000000 00000000 00000000 00000111

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700
```

 /THIS CODE WILL ALWAYS BE LOADED TO FIELD 0 FOR INTERRUPT PROCESSING
 /PURPOSES, REGARDLESS OF WHAT FIELD THE PROGRAM IS LOADED INTO.

```

0000 0000 *0
0000 7402 HLT
0001 6244 RMP
0002 5023 JMP SKPCHN

0142 0142 *PNRADD
0143 1076 PNRUP, TAD SAVHQ
0144 7421 MQL
0145 1075 TAD SAVFLG
0146 6005 RTP
0147 7200 CLA
0148 1154 TAD PPRTWO
0149 7640 SEA CLA
0150 5141 JMP THOBK
0151 5141 TAD SAVAC
0152 1074 JMP I SAVRET
0153 5477
0154 0100 PPRTWO, PARTWO
S
  
```

```

0000 11100000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0100 00000000 00000000 00000000 00000000 00111111 11111000 00000000 00000000
  
```

0200
0300

0400
0500

0600
0700

1000
1100

1200
1300

1400
1500

1600
1700

2000
2100

2200
2300

2400
2500

2600
2700

3000
3100

3200
3300

3400
3500

3600
3700

4000
4100

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

ACDATA 0107	D0MORE 4007	K2 0042	M4200 2705
ACL 7701	DOSIMU 1461	K20 0045	M5 3743
ACLTST 0725	DOSWAP 1570	K200 0051	M5B 3371
ACTINT 2220	EXHLT1 0221	K2000 0055	MQA 7501
ACWAS 0101	FINALE 3713	K201 3027	MQDATA 0110
ADDRER 0450	FSETER 2344	K220 3372	ML 7421
AIBIT3 1140	GDREU 3056	K221 3197	MLTST 0644
AIBIT4 1156	GETEXP 1651	K2525 0056	MQWAS 0102
AIBIT5 1150	GETFND 1660	K2527 3100	NXTONE 1340
AIIDTS 1126	GRISIM 1467	K2552 0057	NXTONN 1743
AINITS 1120	GR2GR3 1507	K3000 2736	NXTPAT 1360
ANDSTS 1032	GROUP2 1511	K3333 0060	NXTPTT 1742
ANISTS 1106	GROUP3 1554	K4 0043	OUT 1527
ANIZTS 1054	GTF 6004	K40 0046	PARTWO 0100
ASYNCOB 3205	GTFTS1 1170	K400 0052	PAVFLG 0136
BACKAD 2653	GTFTS2 1215	K4000 0061	PAVRET 0140
BAKAD9 2706	GTFTS3 2326	K410 2353	PCTST2 2516
BAKADC 2737	HLT 7402	K4201 3742	POINT 0156
BAKADD 3000	HLTCOD 2532	K440 2704	POINTB 0115
BIT11 0112	HWRDES 0021	K442 2702	POINTC 0116
BIT3 0003	INSTRT 1372	K444 2567	POINTD 0117
BIT4 0004	INSTIR 0122	K4444 0062	POINTE 0120
BIT5 0005	INTAOD 2425	K446 2570	POINTF 0121
BIT6 0117	INTEN1 1255	K5000 2703	POINTR 0113
BIT7 0007	IOF 6002	K5225 0063	POINTX 0173
BIT8 0006	IOFHLT 2311	K5252 0064	POINTXA 0174
BKFLOW 3130	IOFTS1 1247	K5253 0065	PQWRON 0125
BKIN17 3146	ION 6001	K6000 2355	PPRTWO 0154
BKINTO 3200	IONTS1 1240	K6666 0066	PSAVNQ 0137
BSW 7002	ISOSTS 1021	K7000 0034	PWRADD 0142
BSWDRP 0623	ISISTS 1102	K7600 1366	PWRUP 0142
BSWPCK 0633	ISIZTS 1052	K77 0047	RARTST 0531
CAF 6007	ISZETS 0757	K7700 0067	RDF 6214
CAM 7621	IZITTS 2503	K7721 0070	READA 6141
CAMTST 0712	JMPTS1 1301	K7770 0071	RETINS 0073
CANCON 0036	JMPTS2 1320	K7777 0072	RETURN 2531
CKSWIT 0124	JMSLO2 1326	KOPHRE 3736	REVSEN 1540
CLRAL 6145	JMSLOC 1311	KCPSTO 3733	RIF 6224
CNTENA 6140	JMST51 1307	KHALT 3665	ROBHLT 3025
CPHRET 3703	JMST52 1323	KJMPIO 3734	ROBRET 3030
CPSTOR 3701	JMST53 2230	KJMPI4 2354	ROMB15 3013
DATPAT 0177	K0 0040	KJMPI7 3735	ROMCLR 3054
DBOUT 3066	K1 0041	KSTOP 0135	ROMH04 2471
DCAZTS 0737	K10 0044	KSTOVR 3737	ROMRTN 3400
DCOSTS 1007	K100 0050	LEAVE 3732	ROMTST 1336
DCISTS 1072	K1000 0131	LKDATA 0111	RTF 0005
DCIZTS 1044	K101 2424	LKAS 0103	RTFTS1 1207
DPBE8A 2451	K1111 0053	LQADA 0146	RTFTS2 1262
DISERR 1646	K12 3271	LQADB 0147	RTLDRP 0006
DLYHLT 2413	K17 3161	LQPAK 1752	RTLPCK 0016
DMARET 3336	K1777 0054	M2 3741	RTRPK 0001
DOAGAN 1670	K17B 3373	M200 2766	SAVAC 0074